



# Anchor Center For Certification (ACC)

July 25, 2025

Regulatory Management Branch

Division of Food Ingredients

Office of Pre-Market Additive Safety

Office of Food Chemical Safety, Dietary Supplements, and Innovation

Human Foods Program

U.S. Food and Drug Administration

5001 Campus Drive HFS-225 College Park, MD 20740

Attention: Kaiping Deng, Ph.D.

Re: GRN 001255 – Response to Questions Posed in an Email on Jul 14, 2025

Dear Dr Deng:

Per your request, Anchor Center for Certification (ACC), acting as the agent for MoreMeat (Guangzhou) Biotech Co., Ltd (MoreMeat), is providing a response to FDA's questions as denoted in your email dated July 14, 2025, as follows:

1. In Table 3, ammonium sulfate is listed as a nitrogen source in the fermentation media, with 21 CFR 184.1143 cited for its regulatory status. However, the regulatory status for ammonium sulfate under 21 CFR 184.1143 applies specifically to its use as a dough strengthener, firming agent, and processing aid in baked goods, gelatins, and puddings. Please provide the correct regulation for your intended use of ammonium sulfate. In addition, please confirm that all raw materials and processing aids used in the production of mycelial biomass of *F. compactum* "MM-135" are food grade and are used in accordance with an appropriate U.S. regulation, are GRAS for their intended use, or are the subject of an effective food contact notification.

## MoreMeat's response:

Ammonium sulfate is used as a nitrogen source in the fermentation medium to support microbial growth and metabolite production (Akpogheli et al., 2024; Yalçın & Özbaş, 2007). While 21 CFR 184.1143 designates ammonium sulfate as a direct food additive for specific functions (dough strengthener, firming agent, processing aid), MoreMeat uses ammonium sulfate as an in-process fermentation nutrient, not as a direct ingredient in the final product.



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In fermentation media, ammonium sulfate's function is to supply bioavailable nitrogen to the production microorganism. Its use in microbial fermentation is well-established in food biotechnology, and it is not expected to be present in the final purified ingredient, thereby aligning with the definition of a processing aid under FDA guidance (e.g., 21 CFR 170.3(o)(24)). Therefore, while the cited CFR section covers direct food use, the use of ammonium sulfate in More Meat's process is functionally and scientifically justified, and consistent with recognized fermentation practices and precedented use in food biotechnology (Papagianni et al., 2005).

Furthermore, there is a prior precedent for the use of ammonium sulfate as a nutrient source in fermentation media used for the production of GRAS ingredients. For example, on November 22, 2024, FDA responded with a "no questions" letter to GRN 001183 for Monellin preparation produced by *Komagataella phaffii* P-MON-40 expressing a gene encoding for a modified monellin, where ammonium sulfate, anhydrous is listed in Table 2 of the GRAS dossier as a "Raw material/processing aid" with the "function/purpose" of "nutrient source."

MoreMeat hereby confirms that that all raw materials, processing aids, and food contact substances used in the production of mycelial biomass of *F. compactum* MM-135 are food grade and are used in accordance with an appropriate U.S. regulation, are GRAS for their intended use, or are the subject of an effective food contact notification.

### References:

- Akpogheli, P. O., Edo, G. I., Kasar, K. A., Zainulabdeen, K., Yousif, E., Mohammed, A. A., Jikah, A. N., Ezekiel, G. O., Owheru, J. O., Ugbune, U., Umar, H., Ekokotu, H. A., Oghoro, E. E. A., Onyibe, P. N., Ekpeko, L. D., Isoje, E. F., & Agbo, J. J. (2024). Impact of different nitrogen sources, initial pH and varying inoculum size on the fermentation potential of *Saccharomyces cerevisiae* on wort obtained from sorghum substrate. *Food Materials Research*, 4, 0. <https://doi.org/10.48130/fmr-0024-0012>
- Papagianni, M., Wayman, F., & Mattey, M. (2005). Fate and Role of Ammonium Ions during Fermentation of Citric Acid by *Aspergillus niger*. *Applied and Environmental Microbiology*, 71(11), 7178–7186. <https://doi.org/10.1128/aem.71.11.7178-7186.2005>
- Yalçın, S. K., & Özbaş, Z. Y. (2007). Effects of ammonium sulphate concentration on growth and glycerol production kinetics of two endogenic wine yeast strain. *Indian Journal of Biotechnology*, 7, 89–93. [https://nopr.niscpr.res.in/bitstream/123456789/2736/1/IJBT%207\(1\)%2089-93.pdf](https://nopr.niscpr.res.in/bitstream/123456789/2736/1/IJBT%207(1)%2089-93.pdf)



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2. Please clarify if the results given in Table 6 are on a wet or dry weight basis. There is no mention in the title, caption, or footnote if these results are wet or dry weight. Additionally, please confirm the moisture content of the article of commerce, since multiple moisture values are reported in various parts of GRN 001255.

### MoreMeat's response:

The results of three non-consecutive lots of MoreMeat's mycoprotein, as shown in Table 6 in the dossier, are reported on a wet weight basis. To further demonstrate the consistency of moisture content presented in the GRAS notification, a summary table is provided below (Table 1). The data confirms that moisture content is consistent across product batches at time zero.

The moisture test results for three non-consecutive lots of MoreMeat's mycoprotein, obtained during the stability study, are summarized in Tables 2 and 3 in response to Question 4. The moisture test results obtained at different timepoints during the stability study are not directly comparable to the time-zero values, as they reflect changes due to storage conditions over time rather than initial product characteristics.

**Table 1 Moisture Content Results from the GRAS notification**

| Test Items                     | Location in the dossier | Test Results      |                   |                    |                    |
|--------------------------------|-------------------------|-------------------|-------------------|--------------------|--------------------|
|                                |                         | Batch #MM13524002 | Batch #MM13524004 | Batch # MM13524007 | Batch # MM13524009 |
| Moisture                       | Table 1 (page 21)       | /                 | 76.3%             | 74.1%              | 75.7%              |
| Moisture (Stability T=0 month) | Table 11 (page 31)※     | 72.5%             | 76.3%             | /                  | 75.7%              |
| Moisture (Stability T=0 month) | Table 12 (page 32)※     | 72.5%             | 76.3%             | /                  | 75.7%              |

Note: ※: Tables 11 and 12 in GRAS dossier present the stability data for MoreMeat's mycoprotein under accelerated and long-term storage conditions, respectively.

/: moisture content was not tested



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3. Please state that all analytical methods are validated for their intended use.

MoreMeat's response:

MoreMeat hereby confirms that all analytical methods are validated for their intended use.

4. You indicate that the stability study was ongoing. Please provide an update on the stability study and indicate if these additional data would change the conclusion that the ingredient is stable up to 12 months when stored at -18 °C.

MoreMeat's response:

The accelerated stability study conducted at 5°C was ongoing at the time of the submission of the notice but has since been completed. The results are summarized in Table 2 below. The long-term stability study conducted at -20°C is still ongoing but MoreMeat is able to provide additional data in Table 3. The additional stability data does not change the conclusion that the ingredient is stable for up to 12 months at -18°C when stored in a vacuum-sealed aluminum bag.



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**Table 2 Stability Test Result for Mycoprotein at Accelerated Conditions (5°C±3°C)**

| Lot #      | Test Month | Appearance                      | Protein<br>(g/100g<br>as wet<br>basis) | Protein<br>(g/100g<br>on dried<br>basis) | Moisture<br>(g/100g) | Total<br>Plate<br>Count<br>(CFU /g) | Yeast &<br>Mold<br>(CFU /g) | <i>L.</i><br><i>monocytogenes</i> | <i>E.</i><br><i>coli</i> | <i>S.</i><br><i>aureus</i> | <i>Salmonella</i> |
|------------|------------|---------------------------------|----------------------------------------|------------------------------------------|----------------------|-------------------------------------|-----------------------------|-----------------------------------|--------------------------|----------------------------|-------------------|
| MM13524002 | 0          | Off-white, dough-like matter    | 13.8                                   | 50.2                                     | 72.5                 | <10                                 | <10                         | ND                                | ND                       | ND                         | ND                |
|            | 1          | Off-white, dough-like matter    | 12.9                                   | 49.0                                     | 73.7                 | <10                                 | <10                         | /                                 | /                        | /                          | /                 |
|            | 2          | Off-white, dough-like matter    | 14.2                                   | 50.0                                     | 71.6                 | <10                                 | <10                         | /                                 | /                        | /                          | /                 |
|            | 3          | Off-white, dough-like matter    | 14.2                                   | 52.6                                     | 73.0                 | <10                                 | <10                         | ND                                | ND                       | ND                         | ND                |
|            | 6          | Off-white, dough-like matter    | 14.5                                   | 53.9                                     | 73.1                 | <10                                 | <10                         | ND                                | ND                       | ND                         | ND                |
| MM13524004 | 0          | light yellow, dough-like matter | 13.2                                   | 55.7                                     | 76.3                 | <10                                 | <10                         | ND                                | ND                       | ND                         | ND                |
|            | 1          | light yellow, dough-like matter | 13.5                                   | 51.1                                     | 73.6                 | <10                                 | <10                         | /                                 | /                        | /                          | /                 |
|            | 2          | light yellow, dough-like matter | 13.2                                   | 50.2                                     | 73.7                 | <10                                 | <10                         | /                                 | /                        | /                          | /                 |
|            | 3          | light yellow, dough-like matter | 13.9                                   | 59.7                                     | 76.7                 | <10                                 | <10                         | ND                                | ND                       | ND                         | ND                |
|            | 6          | light yellow, dough-like matter | 14.5                                   | 59.9                                     | 75.8                 | 320                                 | <10                         | ND                                | ND                       | ND                         | ND                |
| MM13524009 | 0          | light yellow, dough-like matter | 13.2                                   | 54.3                                     | 75.7                 | <10                                 | <10                         | ND                                | ND                       | ND                         | ND                |
|            | 1          | light yellow, dough-like matter | 13.0                                   | 61.3                                     | 77.5                 | <10                                 | <10                         | /                                 | /                        | /                          | /                 |
|            | 2          | light yellow, dough-like matter | 13.0                                   | 54.2                                     | 76.0                 | <10                                 | <10                         | /                                 | /                        | /                          | /                 |
|            | 3          | light yellow, dough-like matter | 12.8                                   | 57.4                                     | 77.7                 | <10                                 | <10                         | ND                                | ND                       | ND                         | ND                |
|            | 6          | light yellow, dough-like matter | 12.9                                   | 57.6                                     | 77.6                 | <10                                 | <10                         | ND                                | ND                       | ND                         | ND                |

ND: not detected, /: not tested



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Table 3 Stability Test Result for Mycoprotein at Long-Term Conditions (-20°C±5°C)

| Lot #      | Test Month | Appearance                      | Protein<br>(g/100 g)<br>as wet<br>basis | Protein<br>(g/100 g)<br>on dried<br>basis | Moisture<br>(g/100 g) | Total<br>Plate<br>Count<br>(CFU /g) | Yeast &<br>Mold<br>(CFU /g) | <i>L.</i><br><i>monocytogenes</i> | <i>E.</i><br><i>coli</i> | <i>S.</i><br><i>aureus</i> | <i>Salmonella</i> |
|------------|------------|---------------------------------|-----------------------------------------|-------------------------------------------|-----------------------|-------------------------------------|-----------------------------|-----------------------------------|--------------------------|----------------------------|-------------------|
| MM13524002 | 0          | Off-white, dough-like matter    | 13.8                                    | 50.2                                      | 72.5                  | <10                                 | <10                         | ND                                | ND                       | ND                         | ND                |
|            | 3          | Off-white, dough-like matter    | 14.0                                    | 52.2                                      | 73.2                  | <10                                 | <10                         | /                                 | /                        | /                          | /                 |
|            | 6          | Off-white, dough-like matter    | 13.9                                    | 51.7                                      | 73.1                  | 10                                  | <10                         | /                                 | /                        | /                          | /                 |
|            | 9          | Off-white, dough-like matter    | 14.2                                    | 51.8                                      | 72.6                  | 40                                  | <10                         | /                                 | /                        | /                          | /                 |
|            | 12         | Off-white, dough-like matter    | 13.4                                    | 50.4                                      | 73.4                  | <10                                 | <10                         | ND                                | ND                       | ND                         | ND                |
| MM13524004 | 0          | light yellow, dough-like matter | 13.2                                    | 55.7                                      | 76.3                  | <10                                 | <10                         | ND                                | ND                       | ND                         | ND                |
|            | 3          | light yellow, dough-like matter | 13.5                                    | 61.6                                      | 78.1                  | <10                                 | <10                         | /                                 | /                        | /                          | /                 |
|            | 6          | light yellow, dough-like matter | 15.4                                    | 56.6                                      | 72.8                  | 10                                  | <10                         | /                                 | /                        | /                          | /                 |
|            | 9          | light yellow, dough-like matter | 13.8                                    | 50.5                                      | 72.7                  | <10                                 | <10                         | /                                 | /                        | /                          | /                 |
|            | 12         | light yellow, dough-like matter | 12.7                                    | 50.4                                      | 74.8                  | <10                                 | <10                         | ND                                | ND                       | ND                         | ND                |
| MM13524009 | 0          | light yellow, dough-like matter | 13.2                                    | 54.3                                      | 75.7                  | <10                                 | <10                         | ND                                | ND                       | ND                         | ND                |
|            | 3          | light yellow, dough-like matter | 13.5                                    | 53.4                                      | 74.7                  | <10                                 | <10                         | /                                 | /                        | /                          | /                 |
|            | 6          | light yellow, dough-like matter | 14.4                                    | 58.1                                      | 75.2                  | 10                                  | <10                         | /                                 | /                        | /                          | /                 |
|            | 9          | light yellow, dough-like matter | 13.4                                    | 50.4                                      | 73.4                  | <10                                 | <10                         | /                                 | /                        | /                          | /                 |

ND: not detected, /: not tested



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5. We noticed discrepancies between Table 13 in GRN 001255 and its Appendix 3 regarding the food categories and use levels for mycelial biomass of *F. compactum* “MM-135”. For example, your uses include processed fruit and fruit juices, but only the food code for orange juice (92550350) is used while there are more than 7 food codes for orange juice in the National Health and Examination Survey (NHANES). We also note that food codes for apple and vegetable juices were omitted. Please clarify these differences, explaining any variations in the food categories between your intended uses and those used in your dietary exposure estimate. Additionally, please address why specific foods were omitted from the dietary exposure assessment when those foods were listed as intended food categories in Table 13.

### MoreMeat’s response:

The GRAS substance mycelial biomass of *Fusarium compactum* MM-135, manufactured by MoreMeat, is intended solely for use in “high-protein juice” products within the categories of Processed Fruits and Fruit Juices and Processed Vegetables and Vegetable Juices, as indicated in Table 13 in the notification. These intended uses do not include 100% fruit or vegetable juices, such as 100% orange juice or 100% apple juice, which would not be expected to contain added protein. Accordingly, product codes representing pure juice products—such as Orange Juice, 100%, NFS (61210000)—have been intentionally omitted from the intake assessment. Instead, the assessment includes product codes most likely to represent high-protein juice formulations, such as 92550360 (Apple juice beverage), along with various vegetable juice blends (e.g., 92550400, 92550405, 75132000, etc.), which more closely align with the intended application of the ingredient.

6. We noted that several sections of your notice contain significant amounts of text copied from GRN 904 and GRN 945. Be aware of the rules to avoid plagiarism; there are guidance documents available, including those provided by the US Government, National Institutes of Health, etc. It is important to note that using quotes and ascribing a source is not enough to avoid plagiarism when the section being copied is long. There is discussion regarding how to discuss data and information incorporated into a GRAS notice on page 54988 of the GRAS Final Rule (62 FR 54988, August 17, 2016).

### MoreMeat’s response:

MoreMeat acknowledges FDA’s observation regarding the textual similarities between our GRAS notice and portions of GRN 000904 and GRN 000945. In preparing our submission, our intent was to accurately convey well-established scientific principles and regulatory concepts relevant to our ingredient, some of which have been previously articulated in publicly available GRAS notices.



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Thank you for your suggestion to review the guidance on how to incorporate data and information from previous GRAS notices into our own GRAS submission. We note that Comment 48 and Response 48 in the 62 FR 54988, Aug 17<sup>th</sup>, 2016 stated that we must have sufficient knowledge of data and information submitted by another party, and to identify the specific source of data and information to be incorporated into our own GRAS submission, and to ensure the information incorporated must be public and not confidential and need to be obtained through FOIA.

We affirm that our GRAS assessment reflects our independent review, and we would like to revise section 6.4 to demonstrate our original scientific evaluation, in alignment with the principles outlined on page 54988 of the GRAS Final Rule (81 FR 54988; August 17, 2016).

## “6.4 Read-Across Rationale and Supporting Evidence

Based on established scientific principles for evaluating the safety of *Fusarium* species, MoreMeat has independently determined that *F. compactum*, *F. strain flavolapis*, and *F. venenatum*, when cultivated under controlled fermentation conditions, demonstrate equivalent toxicological profiles. This conclusion is supported by an integrated assessment of phylogenetic relationships, compositional data, genetic determinants of secondary metabolite production, and analytical testing results. In addition, our conclusion is substantiated by *in vitro* and *in vivo* safety studies specific to MoreMeat’s *F. compactum*-derived mycoprotein.

This “read-across” approach is consistent with scientific principles recognized by the GRAS community, including the use of phylogenetic and compositional comparators to infer toxicological equivalence. Our evaluation aligns with longstanding practices in toxicological risk assessment of biological substances and applies current methodologies in genomics and metabolomics to substantiate the absence of safety concerns.

### 6.4.1 Scientific Basis for Comparative Evaluation

The application of comparative safety assessments has precedent in GRAS determinations involving closely related microbial species. Numerous GRAS notices—such as GRNs 000737, 000811, and 000904—have employed this approach to establish safety based on demonstrable similarities in taxonomy, metabolic activity, and intended use. Although our safety assessment references select data from prior notices, we have conducted an independent and comprehensive evaluation of *F. compactum* using current data and methods.



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We also consider the framework described by Pariza and Johnson (2001) relevant to our assessment. While originally proposed for genetically modified production strains, the broader principles outlined—namely, evaluating an organism’s history of safe use, taxonomic relatedness, and potential for toxin production—remain applicable. With the advancement of high-resolution sequencing and metabolomic profiling, these tools now enable a rigorous comparison across *Fusarium* species.

## 6.4.2 Comparative Toxicological Potential of Key *Fusarium* Species

The toxicological potential of *F. compactum* was assessed relative to *F. strain flavolapis* and *F. venenatum* based on the following key domains:

- **Phylogenetics:** All three species are members of the *Fusarium* genus and fall within closely related clades, indicating high genetic similarity.
- **Composition:** Their respective mycoproteins share similar nutritional, protein, and fiber profiles.
- **Intended Use:** All are intended for use as alternative dietary proteins in similar food applications.
- **Secondary Metabolites:** The profile of mycotoxin-related gene clusters and their expression under controlled growth conditions is consistent across the three strains.
- **Toxicological Data:** All are associated with a low likelihood of mycotoxin production under food-relevant fermentation conditions, supported by targeted analytical testing and toxicology data.

## 6.4.3 Genetic and Analytical Evaluation

Our investigation of *F. compactum* included genomic screening of biosynthetic gene clusters associated with key *Fusarium* mycotoxins (e.g., trichothecenes, enniatins, and beauvericin). The structure and functionality of these gene clusters in *F. compactum* were evaluated in context with homologous sequences in *F. strain flavolapis* and *F. venenatum*, revealing no evidence of increased mycotoxin-producing potential.

These findings were confirmed through targeted analytical testing of fermentation samples, which demonstrated non-detectable or minimal levels of known *Fusarium*-associated mycotoxins under standard production conditions.



## 6.4.4 *In Vitro* and *In Vivo* Safety Studies

To further support the conclusion of safety, MoreMeat conducted a battery of *in vitro* genotoxicity assays (Ames test, micronucleus test, and chromosome aberration test), none of which indicated mutagenic or clastogenic activity. These assays are conducted as per internationally recognized standards (e.g., OECD Test Guidelines) and have been used to assess other *Fusarium*-based mycoproteins.

Additionally, a 14-day oral toxicity study in rats demonstrated no adverse effects at high doses of MoreMeat's mycoprotein. These results are consistent with published data on *F. venenatum*- and *F. strain flavolapis*-derived mycoproteins.

## 6.4.5 Conclusion

The combined genetic, compositional, analytical, and toxicological evidence provides a robust basis to conclude that *F. compactum* poses no greater toxicological risk than *F. strain flavolapis* or *F. venenatum* when grown under controlled production conditions. This conclusion reflects an independent scientific evaluation using contemporary methodologies and is consistent with FDA's expectations for a scientifically grounded and transparent GRAS notice, as outlined in the GRAS Final Rule (81 FR 54988)."

7. On page 63, you state that mycoprotein from *F. compactum*, which is the organism utilized in your notice, has similar nutritional and protein profiles to *F. strain flavolapis* and *F. venenatum*. Please provide evidence to support the comparison of the different *Fusarium* species, such as comparing the metabolome and proteome.

### MoreMeat's response:

In MoreMeat's mycoprotein GRAS notification, MoreMeat states that mycoprotein derived from *F. compactum* exhibits similar phylogenetics, product specification, nutritional and amino acid profile to that of *F. strain flavolapis* and *F. venenatum*. This conclusion is based on:

### **Phylogenetics:**

The phylogenetic analysis presented in Figure 1 demonstrates that *F. compactum* strain MM-135, *F. venenatum*, and *Fusarium strain flavolapis* share a high degree of genetic similarity. Notably, *F.*



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*compactum* MM-135 is more closely related to *F. venenatum*, than *F. strain flavolapis* is to *F. venenatum*. This relationship is further supported by whole genome sequence similarity data: *F. compactum* MM-135 shares approximately 82% genomic similarity with *F. venenatum*, compared to 79% similarity between *F. compactum* MM-135 and *F. strain flavolapis* (Table 4). These findings support the appropriateness of read-across from both comparator species to support the safety evaluation of *F. compactum* MM-135.

**Table 4 Genomic Similarity among *F. compactum* MM-135, *F. venenatum*, and *Fusarium flavolapis***

| % ANI(Coverage)            | <i>F. compactum</i> MM-135 | <i>F. venenatum</i> A 3/5 | <i>F. strain flavolapis</i> MK 7 |
|----------------------------|----------------------------|---------------------------|----------------------------------|
| <i>F. compactum</i> MM-135 | 100(99.96)                 | 82.4957(61.14)            | 79.4448(40.44)                   |

Note: % Average Nucleotide Identity(ANI) among the three strains were calculated using FastANI.

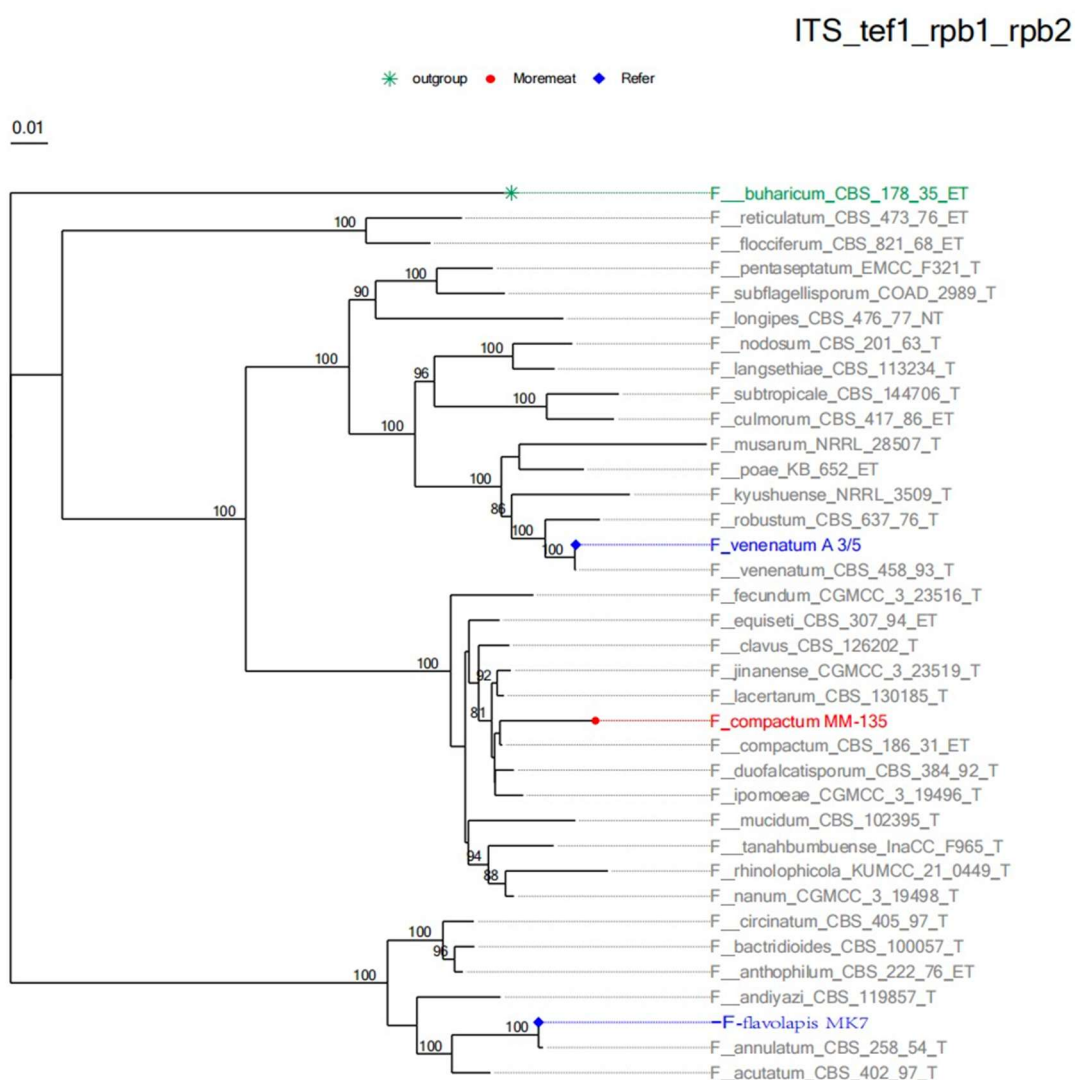


Figure 1 Phylogenetic tree mapping of *F. compactum* MM-135.



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## Product Specification

As shown in Table 28 of the GRAS notification, the product specifications for MoreMeat's mycoprotein are closely aligned with those provided in GRNs 000904 and 000945. While slight differences exist in protein and total dietary fiber levels among the three products, MoreMeat's mycoprotein falls between the other two in both parameters, with differences of less than 1%. These minor variations are well within the expected range for products derived from closely related *Fusarium* species under similar fermentation conditions.

Given the compositional similarity in key macronutrient values—particularly protein and fiber, which are primary nutritional contributors—MoreMeat considers its mycoprotein nutritionally equivalent to the ingredients described in GRNs 000904 and 000945.

MoreMeat concludes that the small differences observed in product specifications do not impact the safety of its mycoprotein ingredient and that its overall nutritional and compositional profile supports the same conclusion of safety as established in GRNs 000904 and 000945.

## Nutritional Profile and Amino Acids

MoreMeat's mycoprotein exhibits a nutritional profile that is comparable to those described in GRNs 000904 and 000945, as presented in Table 33 of this GRAS notification. The protein and dietary fiber content, along with amino acid composition, fall within a similar range, indicating a consistent macronutrient profile across the three *Fusarium*-derived mycoproteins. Moreover, MoreMeat's mycoprotein qualifies as a high-quality protein source, with an amino acid profile comparable to that of beef and pea protein concentrates—both of which are widely accepted in the human diet.

Based on this compositional similarity and established nutritional adequacy, MoreMeat concludes that the overall nutritional profile of its mycoprotein supports the same conclusion of safety as those reached in GRNs 000904 and 000945.

## Mycotoxins

Mycotoxin analysis shows that all three strains have different mycotoxins, but all levels are not toxicologically significant. Mycotoxin analysis (Table 6 of GRAS notification) showed consistent absence or non-toxicologically significant levels of trichothecenes, enniatins, and beauvericin across



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all *Fusarium* three strains when grown under controlled conditions. This suggests a conserved downregulation or absence of mycotoxin biosynthesis under production conditions.

### Conclusion:

When considered alongside the demonstrated similarities in phylogenetic relationship, product specifications, nutritional composition, amino acid profiles, and the absence of toxicologically significant levels of toxic mycotoxins, MoreMeat concludes that the read-across approach from *F. venenatum* and *Fusarium strain flavolapis* is scientifically justified. These comparator species provide a suitable and well-supported basis to form the safety conclusion of MoreMeat's *F. compactum* MM-135 mycoprotein.

8. Based on the specifications provided, there are large variations in the levels of protein and fiber possible in the ingredient. Please provide calculations on potential exposure to protein and fiber using the highest protein level and highest fiber level possible based on specifications for the ingredient and 90<sup>th</sup> percentile dietary exposure. Intake of too much protein or fiber can result in adverse effects. Please discuss the safety of the ingredient and its protein and fiber levels, including Institute of Medicine recommended daily allowances and acceptable macronutrient distribution ranges.

### MoreMeat's response:

The product specification (Table 4 in GRAS notification) for the mycelial biomass of *Fusarium compactum* MM-135 includes protein and total fiber contents, both reported on a dry basis. Based on the analytical results from three non-consecutive lots of MoreMeat's mycelial biomass product, when converted to an as-is basis, these values correspond to a highest protein content of 15% and a total fiber content of 10% (See Table 5 below). MoreMeat has also reviewed additional product batches and confirmed that, when converted to an as-is basis, protein content does not exceed 15% and total fiber content does not exceed 10%.



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**Table 5 Protein and Total Fiber of MoreMeat’s Mycoprotein (Convert to As-is Basis) for Three Non-consecutive Batches**

| Test Items   | Specification for current notification | Test Results      |                    |                    |
|--------------|----------------------------------------|-------------------|--------------------|--------------------|
|              |                                        | Batch #MM13524004 | Batch # MM13524007 | Batch # MM13524009 |
| Moisture※    | ≤78%                                   | 76.3%             | 74.1%              | 75.7%              |
| Protein※     | ≥43%, on dried basis                   | 55.7%             | 57.5%              | 54.3%              |
| Total fiber※ | ≥25.0%, on dried basis                 | 34.0%             | 39.1%              | 38.4%              |
| Protein      | Convert to as-is basis                 | 13.2%             | 14.9%              | 13.2%              |
| Total fiber  | Convert to as-is basis                 | 8.1%              | 10.1%              | 9.3%               |

Note: calculation for as-is protein = protein (on dried basis) × (1-moisture), for example: 55.7%×(1-0.763)=13.2%.

Calculation for as-is total fiber (on dried basis) × (1-moisture), for example: 34.0%×(1-0.763)=8.1%.

※: test results of three non-consecutive batches of product are reported in table 5 (page 21) of GRAS notification.

Accordingly, for the dietary exposure assessments presented in Section 3.3: Estimated Daily Intake of Protein from MoreMeat’s Mycoprotein (page 36) and Section 3.4: Estimated Daily Intake of Fiber from MoreMeat’s Mycoprotein (page 39), the notifier utilized 15% protein and 10% total fiber, expressed on an as-is basis. These values are consistent with, and conservatively reflect, the mean as-is levels calculated from analytical data obtained from three non-consecutive production lots, which yielded average values of 13.75% protein and 9.15% total fiber. The use of rounded, slightly higher values ensures that the estimated daily intakes represent a precautionary upper bound, supporting a robust and conservative safety assessment.

Since the values used in exposure assessments represent the highest protein level and highest fiber level possible based on specifications for the ingredient (on an as-is rather than on a dry weight basis), we respectfully refer the reviewer to Section 3.3 and Section 3.4 for a detailed discussion of the safety of the ingredient, including its protein and fiber content. These sections provide a comprehensive evaluation of potential exposure including Institute of Medicine recommended daily allowances and acceptable macronutrient distribution ranges, and support the conclusion that the intended uses of the mycoprotein are safe under the proposed conditions.



9. On page 71, you discuss consumption of a similar product and the resulting reactions, including vomiting and diarrhea, which could be attributed to the high chitin content. There is no mention of chitin levels in your ingredient or a safety discussion of chitin. Please discuss the chitin content in your ingredient and why this does not pose a safety concern for consumers.

MoreMeat's response:

As noted on page 71 of the GRAS notice, gastrointestinal symptoms such as vomiting or diarrhea have been reported in isolated cases following consumption of certain fungal-based mycoproteins. These effects have been hypothesized to result, in part, from the relatively high content of chitin, a naturally occurring polysaccharide found in the cell walls of filamentous fungi.

### **Chitin Content in MoreMeat's Mycoprotein**

Chitin is an insoluble dietary fiber that is widely recognized as a structural component of fungal cell walls. While chitin is not measured directly in MoreMeat's mycoprotein specification, it contributes significantly to the total dietary fiber content of MoreMeat's mycoprotein, which approximately 10% on a wet weight basis (Table 33 page 74 in the GRAS notification).

It was noted in an EFSA (European Food Safety Authority) panel review in 2022, there is currently no nationally or internationally recognized reference method for the analytical determination of chitin levels (EFSA NDA Panel, 2022).

### **Safety of Chitin at Reported Levels**

Chitin is classified as a non-digestible carbohydrate and is generally recognized as safe when consumed in moderate amounts. It is not absorbed in the human gastrointestinal tract and functions similarly to other insoluble fibers. Several GRAS notices (e.g., GRNs 000091, and page 4 of appendix 5 from 000904) have concluded that chitin at levels typical for fungal mycoproteins does not pose a safety concern for the general population.

Currently, there is no officially established tolerable upper intake level (UL) for chitin in humans set by major regulatory agencies such as the U.S. FDA, EFSA, or JECFA (Joint FAO/WHO Expert Committee on Food Additives). However, its safety and functional role as insoluble dietary fiber are well recognized, particularly when it is part of the fungal biomass or mycoprotein-based foods (Derbyshire et al., 2023).



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Furthermore, human tolerance studies of fungal-derived mycoproteins have not identified chitin as a causative factor for adverse effects at typical intake levels. The total fiber content—including chitin—of MoreMeat’s mycoprotein is similar to that found in mycoproteins already present in the food supply with a history of safe use.

### Consideration of Gastrointestinal Symptoms

The mild gastrointestinal symptoms occasionally reported for fungal proteins have generally been associated with very high intake levels or sudden introduction of high-fiber foods in sensitive individuals (Jacobson & DePorter, 2018). These effects are not unique to chitin and can occur with other high-fiber foods such as legumes, psyllium, or whole grains. The levels of chitin present in MoreMeat’s mycoprotein are not expected to result in adverse effects when consumed under normal dietary patterns.

In conclusion, the chitin content in MoreMeat’s mycoprotein is consistent with that of other fungal mycoproteins with established safety records. Given its classification as a dietary fiber, its known physiological effects, and the absence of evidence of toxicity at the levels consumed, chitin present in MoreMeat’s mycoprotein is concluded to not pose a safety concern under the intended conditions of use. MoreMeat’s conclusion is consistent with the broader scientific understanding of chitin safety and its role as a naturally occurring fiber in fungal-derived ingredients.

### References:

Derbyshire, E. J., Theobald, H., Wall, B. T., & Stephens, F. (2023). Food for our future: the nutritional science behind the sustainable fungal protein – mycoprotein. A symposium review.

*Journal of Nutritional Science*, 12. <https://doi.org/10.1017/jns.2023.29>

EFSA NDA Panel (EFSA Panel on Nutrition, Novel Foods and Food Allergens), Turck D, Bohn T, Castenmiller J, De Henauw S, Hirsch-Ernst KI, Maciuk A, Mangelsdorf I, McArdle HJ, Naska A, Pelaez C, Pentieva K, Siani A, Thies F, Tsabouri S, Vinceti M, Cubadda F, Frenzel T, Heinonen M, Marchelli R, Neuhaus-Berthold M, Poulsen M, Maradona MP, Schlatter JR, van Loveren H, Azzollini D and Knutsen HK, 2022. Scientific Opinion on the safety of partially defatted house cricket (*Acheta domesticus*) powder as a novel food pursuant to Regulation (EU) 2015/2283. EFSA Journal 2022; 20(5):7258, 26 pp. <https://doi.org/10.2903/j.efsa.2022.7258>



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Jacobson, M. F., & DePorter, J. (2018). Self-reported adverse reactions associated with mycoprotein (Quorn-brand) containing foods. *Annals of Allergy Asthma & Immunology*, 120(6), 626–630.  
<https://doi.org/10.1016/j.anai.2018.03.020>

10. You include a discussion on the safety of enniatins and beauvericin found in the ingredient, including calculations based on the average levels measured in the three test batches. Please provide calculations on potential exposure based on the highest level allowed by specifications and 90<sup>th</sup> percentile dietary exposure. In addition, please confirm that the highest potential exposure to enniatins and beauvericin would not pose a safety concern to consumers.

## MoreMeat's response:

Potential exposure calculations based on the highest level allowed by specifications and 90<sup>th</sup> percentile dietary exposure are presented in the following:

## Enniatins:

The product specification of MoreMeat's mycoprotein presented in Table 4 (in the notice) indicate an acceptance criterion for enniatins content is <40 µg/kg. The EDI of enniatins from consumption of MoreMeat's mycoprotein can therefore be calculated for a 90<sup>th</sup> percentile consumer as follows:

$$\frac{40 \mu\text{g}}{\text{kg}} \times \frac{207 \text{ g}}{\text{day}} \times \frac{1 \text{ kg}}{1,000 \text{ g}} = 8.28 \mu\text{g/day}$$

The updated dietary exposure of enniatins based on the highest level allowed by specifications are summarized below:

**Table 6 Enniatins EDIs**

|           | Mean EDI<br>µg/day | 90 <sup>th</sup> percentile<br>EDI, µg/day | Mean EDI,<br>µg/kg bw/day | 90 <sup>th</sup> percentile<br>EDI, µg/kg bw/day |
|-----------|--------------------|--------------------------------------------|---------------------------|--------------------------------------------------|
| Enniatins | 3.76               | 8.28                                       | 0.054                     | 0.118                                            |

Note: Per body weight calculation assumes an individual body weight of 70 kg.

Table 6 provided in this response updates Table 22 in the original dossier.

We have identified no intake estimate of enniatins for U.S. consumers and, given the relative similarity of the European and U.S. diets, we have considered it reasonable to utilize the EU data regarding current intakes of enniatins to be representative of the U.S. population. This approach is utilized in GRN 000904 (page 26) for the dietary intake discussion of mycotoxins in fungal protein derived from *Fusarium strain flavolapis*; FDA had “no questions” regarding about this approach. Therefore, the same approach



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is applied here to evaluate the dietary intake of enniatins from consumption of MoreMeat's mycoprotein. Indeed, as the EU data calculated a 95<sup>th</sup> percentile intake (rather than 90<sup>th</sup> percentile), the use of this data is conservative. We have, therefore, calculated the Cumulative Estimated Daily Intake (CEDI) of enniatins based on the EU intake data in combination with the calculated intakes from MoreMeat's mycoprotein. We first calculated the enniatins EDIs (mean and 90<sup>th</sup> percentile consumer) on a body weight basis assuming an individual body weight of 70 kg. These intakes were added to the upper-bound intake values for the mean and 95% percentile consumers from the CONTAM evaluation to calculate the CEDIs. Thus, the CEDIs are summarized in Table 7 below.

**Table 7 Summary of Enniatins EDI and CEDI**

|           | <b>Mean EDI,<br/>µg/kg<br/>bw/day</b> | <b>90<sup>th</sup><br/>Percentile<br/>EDI,<br/>µg/kg bw/day</b> | <b>CONTAM<br/>Mean<br/>EDI,<br/>µg/kg<br/>bw/day</b> | <b>CONTAM<br/>95<sup>th</sup><br/>Percentile<br/>EDI,<br/>µg/kg<br/>bw/day</b> | <b>CEDI<br/>Mean,<br/>µg/kg<br/>bw/day</b> | <b>CEDI<br/>95<sup>th</sup><br/>Percentile,<br/>µg/kg<br/>bw/day</b> |
|-----------|---------------------------------------|-----------------------------------------------------------------|------------------------------------------------------|--------------------------------------------------------------------------------|--------------------------------------------|----------------------------------------------------------------------|
| Enniatins | 0.054                                 | 0.118                                                           | 0.91                                                 | 3.28                                                                           | 0.964                                      | 3.398                                                                |

Note: Table 7 provided in this response updates Table 23 in the original dossier.

### Beauvericin:

The product specification of MoreMeat's mycoprotein presented in Table 4 (in the notice) indicates an acceptance criterion for beauvericin content is <10 µg/kg. The EDI of beauvericin from consumption of MoreMeat's mycoprotein can therefore be calculated for a 90<sup>th</sup> percentile consumer as follows:

$$\frac{10 \mu\text{g}}{\text{kg}} \times \frac{207 \text{ g}}{\text{day}} \times \frac{1 \text{ kg}}{1,000 \text{ g}} = 2.07 \mu\text{g/day}$$

The updated dietary exposure of beauvericin based on the highest level allowed by specifications are summarized below:

**Table 8 Beauvericin EDI**

|             | <b>Mean EDI,<br/>µg/day</b> | <b>90<sup>th</sup> Percentile<br/>EDI, µg/day</b> | <b>Mean EDI,<br/>µg/kg bw/day</b> | <b>90<sup>th</sup> Percentile<br/>EDI, µg/kg<br/>bw/day</b> |
|-------------|-----------------------------|---------------------------------------------------|-----------------------------------|-------------------------------------------------------------|
| Beauvericin | 0.94                        | 2.07                                              | 0.013                             | 0.030                                                       |

Note: Per body weight calculation assumes an individual body weight of 70 kg



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Table 8 provided in this response updates Table 25 of the original dossier

We have identified no intake estimate of beauvericin for U.S. consumers and given the relative similarity of the European and U.S. diets, we have considered it reasonable to utilize the EU data regarding current intakes of beauvericin to be representative of the U.S. population. This approach is utilized in GRN 000904 (page 26) for the dietary intake discussion of beauvericin; FDA had “no questions” regarding about this approach. Therefore, the same approach is applied here to evaluate the dietary intake of beauvericin from consumption of MoreMeat’s mycoprotein. Indeed, as the EU data calculated a 95<sup>th</sup> percentile intake (rather than 90<sup>th</sup> percentile), the use of this data is conservative. We have, therefore, calculated the CEDI to beauvericin based on the EU intake data in combination with the calculated intakes from MoreMeat’s mycoprotein. We first calculated the beauvericin EDIs (mean and 90<sup>th</sup> percentile consumer) on a body weight basis assuming an individual body weight of 70 kg. These intakes were added to the upper-bound intake values for the mean and 95% percentile consumers from the CONTAM evaluation to calculate the CEDIs. Thus, the CEDIs are summarized in Table 9 below.

**Table 9 Summary of Beauvericin EDI and CEDI**

|             | <b>Mean<br/>EDI,<br/>µg/kg<br/>bw/day</b> | <b>90<sup>th</sup><br/>Percentile<br/>EDI, µg/kg<br/>bw/day</b> | <b>CONTAM<br/>Mean EDI,<br/>µg/kg bw/day</b> | <b>CONTAM<br/>95<sup>th</sup><br/>percentile<br/>EDI,<br/>µg/kg bw/day</b> | <b>CEDI<br/>Mean,<br/>µg/kg<br/>bw/day</b> | <b>CEDI<br/>95<sup>th</sup><br/>Percentile,<br/>µg/kg<br/>bw/day</b> |
|-------------|-------------------------------------------|-----------------------------------------------------------------|----------------------------------------------|----------------------------------------------------------------------------|--------------------------------------------|----------------------------------------------------------------------|
| Beauvericin | 0.013                                     | 0.030                                                           | 0.05                                         | 0.93                                                                       | 0.063                                      | 0.96                                                                 |

Note: Table 9 provided in this response updates table 26 of the original dossier

Conclusion: Updated calculations using the highest levels permitted by the specification demonstrate that the amounts of enniatins and beauvericin introduced by MoreMeat’s mycoprotein represent only a small fraction of the existing dietary exposure. Therefore, the limit of enniatins and beauvericin established by MoreMeat’s product specification ensures that consumer exposure remains consistently within safe levels.

11. In Section 6.4, you state that “*F. compactum*, *F. strain flavolapis*, and *F. venenatum* grown under controlled conditions possess the same toxicological potential or lack thereof.” The ingredient



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that is the subject of your notice from *F. compactum* contains detectable levels of mycotoxins, while there were no detectable levels of mycotoxins from *F. strain flavolapis*, and *F. venenatum* in the referenced GRNs. Please provide discussion on the presence of mycotoxins in your ingredient under the controlled growing conditions. Additionally, please discuss how the three strains possess the same toxicological potential if they result in mycotoxin differences.

### MoreMeat's response:

MoreMeat has compiled a comparison table of mycotoxin levels in its mycoprotein product and those reported in GRN 000904 and GRN 000091 for mycoproteins derived from *F. strain flavolapis* and *F. venenatum*, respectively. Table 10 demonstrates that most mycotoxins were not detectable across all three mycoprotein sources. The only exceptions are enniatins and beauvericin, which are present at low, non-toxicologically significant levels in MoreMeat's mycoprotein. As discussed in response to Question 10, even the maximum levels permitted under MoreMeat's product specification for enniatins and beauvericin contribute only a small fraction to overall dietary exposure. Therefore, MoreMeat considers the specification limits to be appropriately set, ensuring that consumer exposure remains within safe and acceptable levels.

**Table 10 Comparison of mycotoxin levels of MoreMeat's mycoprotein product to mycoprotein in GRN 000904 and GRN 000091**

| Mycotoxins (µg/kg wet weight)         | Mycoprotein derived from <i>F. venenatum</i> <sup>a</sup> | Mycoprotein derived from <i>F. strain flavolapis</i> <sup>b</sup> | MoreMeat's mycoprotein <sup>c</sup> |
|---------------------------------------|-----------------------------------------------------------|-------------------------------------------------------------------|-------------------------------------|
| Aflatoxin B1                          | Not reported                                              | Not detectable                                                    | Not reported                        |
| Total aflatoxins (Sum B1, B2, G1, G2) | Not reported                                              | Not detectable                                                    | Not detectable                      |
| Ochratoxin A                          | Not reported                                              | Not detectable                                                    | Not detectable                      |
| HT2 & T2                              | Not reported                                              | Not detectable                                                    | Not detectable                      |
| Deoxynivalenol (DON)                  | Not detectable                                            | Not detectable                                                    | Not detectable                      |
| 3-acetyl-DON                          | Not detectable                                            | Not reported                                                      | Not detectable                      |
| Nivalenol                             | Not detectable                                            | Not detectable                                                    | Not detectable                      |



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| Mycotoxins (µg/kg wet weight)     | Mycoprotein derived from <i>F. venenatum</i> <sup>a</sup> | Mycoprotein derived from <i>F. strain flavolapis</i> <sup>b</sup> | MoreMeat's mycoprotein <sup>c</sup> |
|-----------------------------------|-----------------------------------------------------------|-------------------------------------------------------------------|-------------------------------------|
| Diacetoxyscirpenol                | Not detectable                                            | Not detectable                                                    | Not detectable                      |
| Fusarenon X                       | Not detectable                                            | Not detectable                                                    | Not detectable                      |
| Neosolaniol                       | Not detectable                                            | Not detectable                                                    | Not detectable                      |
| 15-acetyl-DON                     | Not reported                                              | Not reported                                                      | Not detectable                      |
| Fumonisin B1                      | Not reported                                              | Not reported                                                      | Not reported                        |
| Fumonisin B2                      | Not reported                                              | Not reported                                                      | Not reported                        |
| Fumonisin (sum of B1, B2 and B3)  | Not reported                                              | <0.5 ppm*                                                         | Not detectable                      |
| Zearalenone                       | Not reported                                              | Not detectable                                                    | Not detectable                      |
| Enniatin (sum of A, A1, B and B1) | Not reported                                              | Not reported                                                      | <40 ppb                             |
| Beauvericin                       | Not reported                                              | <0.1 ppm*                                                         | <10 ppb                             |
| Fusaric acid                      | Not reported                                              | Not reported                                                      | Not detectable                      |
| Moniliformin                      | Not reported                                              | Not reported                                                      | Not detectable                      |
| Fusarin C                         | Not reported                                              | Not reported                                                      | Not detectable                      |
| Fusaproliferin                    | Not reported                                              | Not reported                                                      | Not detectable                      |
| Cyclonerodiol                     | Not reported                                              | Not reported                                                      | Not detectable                      |

Note: <sup>a</sup>The mycotoxin data for mycoproteins derived from *F. venenatum* is reported on page 19-20 of GRN 000091.

<sup>b</sup>The mycotoxin data for mycoproteins derived from *F. strain flavolapis* is reported on Table 23 (page 46) of GRN 000904.

※Specification limit of Fumonisin and Beauvericin are reported on Table 4 (page 13) of GRN 000904

<sup>c</sup>The mycotoxin levels for MoreMeat's ingredient are based on the data presented in Table 6 of its GRAS notification, with the exception of enniatins and beauvericin, for which the product specification limits are used.



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Additionally, MoreMeat have conducted a suite of toxicological studies—including genotoxicity assays and a 14-day *in vivo* toxicity study—on their mycoprotein. These studies demonstrated no adverse effects at the highest doses tested, indicating that the low levels of mycotoxins present do not result in toxicological concern. Therefore, MoreMeat considers the overall toxicological potential of the *F. compactum* MM-135 mycoprotein ingredient comparable to that of the other *Fusarium*-derived mycoproteins.

In conclusion, while there are differences in detected mycotoxin levels (enniatins and beauvericin), the consistent absence of adverse effects in toxicological studies and the maintenance of mycotoxin levels well below safety thresholds support MoreMeat's conclusion that *F. compactum* MM-135 has a comparable toxicological profile to *F. strain flavolapis* and *F. venenatum* under intended use.

12. You reference a previous *in vitro* digestibility study on mycoproteins (Goodman, 2019), but this study could not be located based on the reference information provided.

MoreMeat's response:

The *in vitro* digestibility study on mycoproteins is attached in this response letter as Appendix 1.

13. A preliminary 14-day toxicity study discussed in Section 6.2.2 appears to have been published in May 2025 (Xian et al., 2025) after submission of the GRAS notice. The published study states there are ongoing 90-day studies. Toxicological studies are pivotal in determining the safety of consumption of food ingredients, including the maximum safe level of exposure. Such pivotal safety information should be publicly available before the GRAS notice is submitted to FDA. Please discuss why the potential results of this study are not necessary to demonstrate long term safety of the ingredient. Data or information presented in a GRAS submission should be publicly available and generally accepted, such as those published in reputable peer-reviewed journals and other reputable sources. Such data or information should be made available in the public domain for a few months to allow for public commentary.

MoreMeat's response:

MoreMeat appreciates FDA's emphasis on the importance of publicly available and generally recognized scientific data in support of a GRAS conclusion. In MoreMeat's mycoprotein GRAS notification, the pivotal toxicological studies used to support the safety of the ingredient is a battery of toxicity studies originally submitted and discussed in GRN 000091 and GRN 000904 (Section 6.3 of



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GRAS notification). As detailed in Section 6.4 of MoreMeat's notification, we determined that these studies are directly applicable to MoreMeat's ingredient due to similarities in phylogenetic relationship, product specifications, nutritional composition, amino acid profiles, and the absence of impactful levels of toxic secondary metabolites between MoreMeat's mycoprotein and that described in GRN 000091 and GRN 000904. Therefore, we consider the toxicological studies from GRN 000091 and GRN 000904 to be the appropriate primary source of safety data for our GRAS conclusion.

Importantly, GRN 000091 (Miller & Dwyer, 2001) and GRN 000904 (Furey et al., 2022) made the relevant toxicological studies publicly available prior to our GRAS notice submission. These studies were published in peer-reviewed journals and formed part of the publicly accessible body of scientific literature, allowing for sufficient time for public scrutiny and commentary. By relying on this publicly available data, MoreMeat's GRAS notification complies with FDA's expectation that GRAS determinations be based on information that is accessible in the public domain and generally recognized by qualified experts.

While the more recent 14-day study (Xian et al., 2025), referenced in Section 6.2.2, serves as supportive evidence for MoreMeat's safety conclusion, the pivotal toxicological evidence remains the published and widely accepted studies from GRN 000091 and GRN 000904.

In summary, MoreMeat's GRAS conclusion is grounded in already published, peer-reviewed, and publicly available toxicological studies that are scientifically robust and generally recognized. Additionally, the results of the more recent studies are not pivotal to support the long-term safety of the ingredient under its intended conditions of use.

### References:

- Miller, S. A., & Dwyer, J. T. (2001). Evaluating the Safety and Nutritional Value of Mycoprotein. *Food Technology*, 55(7), 42–47. <https://www.ift.org/news-and-publications/food-technology-magazine/issues/2001/july/features/developing-food-safety>
- Furey, B., Slingerland, K., Bauter, M. R., Dunn, C., Goodman, R. E., & Koo, S. (2022). Safety evaluation of Fy Protein™ (Nutritional Fungi Protein), a macroingredient for human consumption. *Food and Chemical Toxicology*, 166, 113005. <https://doi.org/10.1016/j.fct.2022.113005>



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14. Please perform an updated literature search and confirm whether you have retrieved/uncovered any data and information since you submitted the notice that could contradict your GRAS conclusion. Please provide the month and year of when the updated literature search is conducted.

### MoreMeat's response:

An updated literature search was conducted on July, 2025 by MoreMeat. To obtain a current, thorough, and comprehensive understanding of publicly available studies that support the safe use of mycoprotein, literature searches were conducted using both PubMed and Google Scholar to identify published studies. The search terms “mycoprotein”, “fungal protein”, “mycoprotein AND genotoxicity”, “fungal protein AND genotoxicity”, “mycoprotein AND safety”, “fungal protein AND safety” were used to identify recently published studies conducted with mycoprotein since Feb 2025.

No new publications addressing the safety of mycoprotein were identified, other than the study conducted on MoreMeat's mycoprotein (Xian et al., 2025).

### References:

Xian, Y., Yin, L., Wu, Z., Chen, D., Sun, Z., Jiang, X., & Tian, L. (2025). Characterization and preliminary safety evaluation of mycoprotein from *Fusarium compactum* MM-135. *Frontiers in Sustainable Food Systems*, 9. <https://doi.org/10.3389/fsufs.2025.1583642>

15. You state that “The genome of the *F. compactum* MM-135 production strain was sequenced by MoreMeat using an Illumina NextSeq 2000 Sequencer. The analysis indicated that the genome could express 12,548 proteins based on gene prediction and annotation of the DNA.” (page 67).
- For the administrative record, please clarify whether genome data of the *F. compactum* “MM-135” is available in the public domain, e.g., a NCBI accession number.
  - On page 11, you state that “.....that *F. compactum* MM-135 (production organism of the current notification), *F. venenatum* (production organism of GRN 000945 and GRN 000091), and *F. strain flavolapis* (production organism of GRN 000904) belong to neighboring groups of *Fusarium* species. Therefore, they have close evolutionary relationship, sharing large genetic similarities.” Please provide the percentage of genomic



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similarity among *F. compactum* “MM-135”, *F. venenatum*, and *Fusarium flavolapis*.

### MoreMeat’s response:

- a. The genome data of the *F. compactum* MM-135 is not available in the public domain. However, the organism *F. compactum* MM-135 is deposited in the China General Microbiological Culture Collection Center (CGMCC) with the designation number of 41312.
- b. The following table presents the percentage of genomic similarity among *F. compactum* MM-135, *F. venenatum*, and *Fusarium strain flavolapis*. MoreMeat confirms that these three strains share a close evolutionary relationship, as evidenced by their high degree of genomic similarity.

**Table 11 Genomic Similarity among *F. compactum* MM-135, *F. venenatum*, and *Fusarium flavolapis***

| % ANI(Coverage)            | <i>F. compactum</i> MM-135 | <i>F. venenatum</i> A 3/5 | <i>F. strain flavolapis</i> MK 7 |
|----------------------------|----------------------------|---------------------------|----------------------------------|
| <i>F. compactum</i> MM-135 | 100(99.96)                 | 82.4957(61.14)            | 79.4448(40.44)                   |

16. In Part 2.1 (page 12), you state that “MoreMeat confirms that *F. compactum* is non-pathogenic and non-toxicogenic.” For the administrative record, please confirm that the production strain *F. compactum* “MM-135” is non-pathogenic and non-toxicogenic.

### MoreMeat’s response:

MoreMeat confirms that the production strain *F. compactum* MM-135 is non-pathogenic and non-toxicogenic.

17. In part 2.2 (page 12), you state that “the production organism is not genetically modified and there are no antibiotics added in the fermentation process.” For the administrative record, please include a statement to confirm whether the production strain *F. compactum* “MM-135” carries any antibiotic-resistant genes.

### MoreMeat’s response:

Genome-wide resistance gene prediction analysis of MoreMeat’s production organism, *F. compactum* MM-135, confirmed the absence of clinically relevant antimicrobial resistance (AMR) genes, including those prioritized by the World Health Organization (WHO). This indicates a low risk associated with the presence of transferable resistance traits.



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Furthermore, *F. compactum* MM-135 is utilized exclusively in a closed fermentation system and is subject to an inactivation step during downstream processing. These measures significantly reduce the likelihood of environmental exposure or horizontal gene transfer to other microorganisms. Therefore, the use of *F. compactum* MM-135 in the production of mycoprotein does not pose a meaningful risk with respect to antimicrobial resistance.

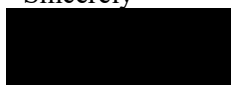
18. Please confirm that fermentation process is performed under a controlled and sterile condition, and is continuously monitored for contaminants.

### MoreMeat's response:

MoreMeat hereby confirms that fermentation process is performed under a controlled and sterile condition, and is continuously monitored for contaminants.

If additional information or clarification is needed as you and your colleagues proceed with the review, please feel free to contact me via email.

Sincerely



Stella Si,

Managing Director of Anchor Center for Certification

No. 1295 Chuan Qiao Road, Building 2, Suite 302,

Pudong, Shanghai, China 201206

[stella.si@acclab.cn](mailto:stella.si@acclab.cn)

# **Report On The In Vitro Digestibility Of *Fusarium compactum***

MoreMeat (Guangzhou) Biotech Co.,Ltd.

October 28, 2024

Address: Room 101, Building C1, No. 11, Kaiyuan Avenue, Huang-p  
u District, Guangzhou City, Guangdong Province, China

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**Abbreviations**

|          |                                                                                                                                                                      |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SGF      | Simulated gastric fluid (without pepsin)                                                                                                                             |
| E-SGF    | Simulated gastric fluid (with pepsin)                                                                                                                                |
| SIF      | Simulated intestinal fluid (without pancreatin)                                                                                                                      |
| E-SIF    | Simulated intestinal fluid (with pancreatin)                                                                                                                         |
| Pmx/Pmix | Protein without pepsin / Protein without pancreatin                                                                                                                  |
| Dmx/Dmix | Digestion sample mixture protein (test protein) plus pepsin / Digestion sample mixture protein (test protein or control protein) plus pancreatin                     |
| Emx/Emix | Enzyme mixture (with pepsin, without protein) / Enzyme mixture (with pancreatin, without protein)                                                                    |
| D0-120   | Digestion samples (target <i>Fusarium compactum</i> protein extracts or RUBISCO plus pepsin or pancreatin) from time 0 (quenched prior to digestion) to time 120 min |
| P1/10    | Experimental control protein at 10% loading                                                                                                                          |
| P0       | Experimental control protein without enzyme, time 0                                                                                                                  |
| P60      | Experimental control protein without enzyme, 60 min                                                                                                                  |
| P120     | Experimental control protein without enzyme, 120 min                                                                                                                 |
| SDS      | Sodium dodecyl sulfate                                                                                                                                               |
| Tris     | Tris (hydroxymethyl) aminomethane                                                                                                                                    |
| w/v      | Solute weight to solution volume                                                                                                                                     |
| v/v      | Solute volume to solution volume                                                                                                                                     |
| PLB      | Protein loading buffer                                                                                                                                               |
| FC       | <i>Fusarium compactum</i>                                                                                                                                            |
| PAGE     | Polyacrylamide gel electrophoresis                                                                                                                                   |

# MoreMeat (Guangzhou) Biotech Co.,Ltd.

## Introduction

MoreMeat (Guangzhou) Biotech Co.,Ltd. has developed a novel food product based on a new *Fusarium* species (*Fusarium compactum*, MM-135), which is obtained through a strict fermentation process. The company is currently declaring a new food ingredient to the regulatory authorities and requires a safety evaluation of the new ingredient, with protein allergenicity being a key assessment.

The Codex Alimentarius Commission guidelines for assessing the allergenicity of novel foods (2003) recommends assessing the novel food ingredient for stability in pepsin at acidic pH using standard conditions as an assay to help evaluate whether the novel ingredient is likely to increase the rate of sensitization or increase the likelihood of eliciting an allergic response in consumers. In addition, the stability of mycoprotein was tested using simulated intestinal fluid digestion experiments, which the Codex Alimentarius Commission recognizes as a useful method to consider potential allergic risks when conducting safety assessments of foods with new protein sources. In vitro digestion provides a simple reference basis to assess protein digestion rates and the potential increase in food allergy risks. Pepsin has optimal activity between pH 1.2 and 2.0, but its activity significantly decreases at pH 3.5 and becomes irreversibly denatured at pH 7.0. Thomas et al. (2004) found that protein digestion at pH 2.0 compared to pH 1.2 results in slightly slower degradation of full-length proteins and protein fragments, without changing the overall sensitivity of proteins to digestion. Therefore, in this study we only assessed the stability of the test protein in simulated gastric fluid at pH 2.0, with auxiliary assessments in simulated intestinal fluid at pH 7.5.

## Part 1: The In Vitro Digestion of Protein Extracts (FC) using E-SGF

### 1. Reagents and Instruments

1. Pepsin quenching solution: 200 mM NaHCO<sub>3</sub>, pH 11 (0.84g NaHCO<sub>3</sub> into 30 ml of water, adjusted pH to 11 with 2 N NaOH, made the final volume to 50mL);
2. 5× SDS-PAGE Protein Loading Buffer (PLB): YEASEN #20315ES;
3. SGF (pH 2.0): 122.8 mg NaCl into 59.84mL water (35 mM NaCl); then, adjusted pH to 2.0 with 160μL of 6 N HCl;
4. E-SGF (pH 2.0): 26 mg pepsin into 2 mL of prepared SGF (pH 2.0); Pepsin (Sigma # P7012, laboratory-determined enzyme activity 2350U/mg, would need 13mg/mL);
5. 15-well 4-20% High-Resolution Precast Gel (Bis-Tris): YEASEN, 36270ES10;
6. Gel Staining Solution: Dissolved 1.25g Coomassie Brilliant Blue R-250 in 225mL methanol, 50mL acetic acid and 225mL water;
7. Destaining solution for Coomassie Brilliant Blue R-250: H<sub>2</sub>O, methanol, and acetic acid in a ratio of 65/25/10 (v/v/v);
8. RealBand Pre-stained Protein Marker, 3 colors, Wide Range, 3.5~235 kDa for tris-mops running buffer, Sangon Biotech, cat#C610016;
9. Bio-Rad PowerPac Basic Power Pack;

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10. ChampGel 5000 Gel Image Analyzer;
11. HWS-26 Type Electric Heating Water Bath: Shanghai Yiheng Scientific Instrument Co., Ltd.;
12. SY-1230 Constant Temperature Water Bath: Suzhou Jiemei Electronics Co., Ltd.;
13. Gen5 ELISA reader: BioTek, USA;
14. HC-3016R High-Speed Refrigerated Centrifuge: Anhui Zhongke Zhongjia Scientific Instruments Co., Ltd.

## 2. Experimental Methods

### 2.1 Sample Preparation

Samples were prepared by grinding frozen dry pellets of sample *Fusarium compactum* (Moremeat, Batch# MM13524002) with approximately 1 g in mass in clean mortar and pestles. Sample extraction was performed using 1:3 protein: buffer ratio with 1 g *Fusarium*, extracted with 2985  $\mu\text{L}$  PBS with 15  $\mu\text{L}$  Triton X 100 to solubilize proteins. The mixture was shaken for 2 hours at room temperature, then centrifuged for 30 minutes at 12000 x g at 4 °C to clarify the solution. BCA method was used to determine the protein concentration as 13.25 mg/mL. Samples were stored at -80°C after dispensing.

### 2.2 Procedure

- (1) Preheated two water baths to  $37 \pm 0.1$  °C and  $85 \pm 0.3$  °C;
- (2) Prepared SGF and E-SGF, incubated in the water bath. Pepsin activity and the amount of tested protein correspond to 10 U/ $\mu\text{g}$  in the digestion solution.

### 2.3 Reaction Tubes

- (1) Labeled three 2.0 ml centrifuge tubes as Pmx, Emx, Dmx;
- (2) Thirteen 1.5mL tubes were labeled as P1/5, P0, P60, D0, D0.5, D2, D5, D10, D20, D30, D60, E0, E60.
- (3) 35 $\mu\text{L}$  pepsin quenching solution (200 mM  $\text{NaHCO}_3$ , pH 11) and 35 $\mu\text{L}$  PLB were added into each labeled tube in point 2.

### 2.4 Protein Solution

- (1) P tube: Pipetted an appropriate amount of protein extraction after thawing to room temperature and labeled as P (13.25mg/ml);
- (2) Pd tube: Made a 1 in 5 dilution of P and labeled as Pd (10 $\mu\text{L}$  P + 40 $\mu\text{L}$  water);

### 2.5 Preparation of digestion, protein, and enzyme controls.

- (1) Protein solution, pepsin solution, and water were preheated separately for 5 minutes at 37°C before the start of each experiment;
- (2) P1/5: Added 81  $\mu\text{L}$  E-SGF, quickly heated to 85°C for 10 minutes, and added 19  $\mu\text{L}$  of protein Pd, mixed well, heated at 85°C for 10 minutes, and stored on ice;
- (3) D0: Added 81  $\mu\text{L}$  E-SGF, quickly heated to 85°C for 10 minutes, pipette 19  $\mu\text{L}$  of protein solution (P), mixed well, heated at 85°C for 10 minutes, and stored on ice;

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- (4) Pmx (protein control without pepsin): Pipetted 175μL of P tube, added 0.76mL SGF and mixed well;

P0: Pipetted 100μL from Pmx, mixed well, heated at 85°C for 10 minutes, and stored on ice;

P60: Heated the remaining Pmx at 37°C for 60 minutes, pipetted 100μL into P60, mixed well, heated at 85°C for 10 minutes, and stored on ice;

- (5) Emx (pepsin control without protein): Pipetted 175μL of water, added 0.76mL E-SGF and mixed well;

E0: Pipetted 100μL from Emx, mixed well, heated at 85°C for 10 minutes, and stored on ice;

E60: Heated the remaining Emx at 37°C for 60 minutes, pipetted 100μL into E60, mixed well, heated at 85°C for 10 minutes, and stored on ice;

- (6) Dmx (digestion mixture): Pipetted 175μL from the P tube, added 0.76mL E-SGF and mixed well, then placed Dmx in 37°C water bath, pipetted 100μL of the digestion mixture at 0.5, 2, 5, 10, 20, 30, 60 minutes into D0.5-D60 tubes respectively, immediately heated each tube to 85°C for 10 minutes, and stored on ice.

### 2.6 Electrophoresis conditions

|          |           |            |           |             |           |           |            |            |            |            |             |           |            |          |
|----------|-----------|------------|-----------|-------------|-----------|-----------|------------|------------|------------|------------|-------------|-----------|------------|----------|
| 1        | 2         | 3          | 4         | 5           | 6         | 7         | 8          | 9          | 10         | 11         | 12          | 13        | 14         | 15       |
| <b>M</b> | <b>P0</b> | <b>P60</b> | <b>D0</b> | <b>D0.5</b> | <b>D2</b> | <b>D5</b> | <b>D10</b> | <b>D20</b> | <b>D30</b> | <b>D60</b> | <b>P1/5</b> | <b>E0</b> | <b>E60</b> | <b>M</b> |

(Sample loading volume: 13 μL/well)

Type of Gel: 4-20% High-Resolution Pre-cast Gel (Bis-Tris);

Marker (M): 3.5-235 KDa;

Running Buffer: Tris-MOPS-SDS Running Buffer Powder, 1 ×;

Electrophoresis conditions: 120V, about 60-80 minutes;

Fixing, staining, and destaining procedure: Staining with Coomassie Brilliant Blue R-250 staining solution for 10 min, decolorization overnight (at least 12 hours), then photographed using gel imaging system.

| Lane | Description                           | Incubation time |     |
|------|---------------------------------------|-----------------|-----|
| M    | Marker (3.5-235 kDa)                  | NA              |     |
| 1    | protein control without pepsin, (P0)  | 0               | min |
| 2    | protein control without pepsin, (P60) | 60              | min |
| 3    | protein in E-SGF, (D0)                | 0               | min |
| 4    | protein in E-SGF, (D0.5)              | 0.5             | min |
| 5    | protein in E-SGF, (D2)                | 2               | min |
| 6    | protein in E-SGF, (D5)                | 5               | min |

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|    |                                          |    |     |
|----|------------------------------------------|----|-----|
| 7  | protein in E-SGF, (D10)                  | 10 | min |
| 8  | protein in E-SGF, (D20)                  | 20 | min |
| 9  | protein in E-SGF, (D30)                  | 30 | min |
| 10 | protein in E-SGF, (D60)                  | 60 | min |
| 11 | 20% protein with quenched pepsin, (P1/5) | 0  | min |
| 12 | Experimental control pepsin, (E0)        | 0  | min |
| 13 | Experimental control pepsin, (E60)       | 60 | min |
| M  | Marker (3.5-235 kDa)                     | NA |     |

### 2.7 Limit of Detection

The protein extract sample was diluted and mixed with the loading buffer, pipetted 20 $\mu$ g as 100% digested sample loading volume, 2 $\mu$ g as 10% digested sample loading volume, the electrophoresis conditions were the same as 2.6.

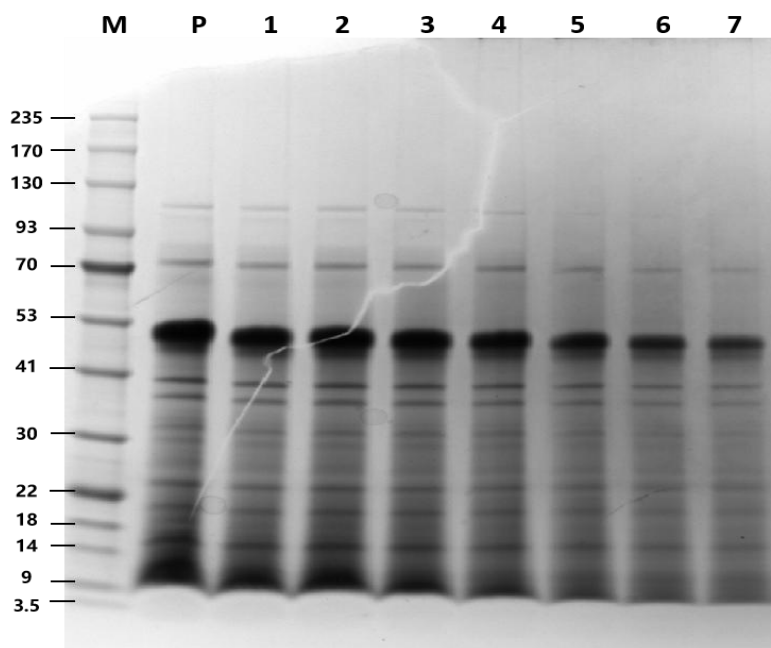
## 3. Results

### 3.1 Limit of Detection.

Figure 1. illustrates that all protein (different molecular weight) bands could be shown on the gel when the protein loaded amount was  $\geq 4$   $\mu$ g/well (lanes P and 1-6), while the largest molecular weight protein band ( $\sim 120$  kDa) could no longer be displayed in lane 7 ( $< 4$   $\mu$ g). Therefore, we concluded that the minimum amount of protein that was reliably detected (limit of detection) was 4  $\mu$ g sample protein, which ensure that all bands could be shown in the gel. Based on these data, the limit of detection was approximately 20% (1/5) of the sample protein loaded in digestion samples, 20 $\mu$ g/well was taken as 100% of loading digestion samples.

The protein loading mentioned here refers to the loading of total proteins, which contains a wide range of proteins, especially the relatively low proportion of high molecular weight proteins. The initial protein concentration of sample extract was 13.25 mg/mL based on BCA method. If we set the digestion endpoint at 10% protein residue, 1/10 sample dilution and other reaction solutions adding would result in lower concentration of total protein. Further dilution may cause the total proteins bands could not be fully shown in the gel, especially high molecular weight protein (like 120 kDa).

Based on the result, we set the end of simulated gastric digestion at 20% protein residue (1/5 of sample protein). The P1/5 protein solution was seen as the end reference for evaluating digestion stability in SGF.



**Figure 1. SDS-PAGE Gel showing the serial dilution of *Fusarium compactum* proteins starting from 100% of total protein.** Lanes 1-7 represent diluted extracts ranging from 40µg (lane 1) to 2µg (lane 7). Lane M and P represent the marker and the protein extract original solution.

### 3.2 SGF Digestion results for *Fusarium compactum*

The electrophoresis result of the simulated gastric fluid digestion test sample was shown in Figure 2., The tested protein and pepsin themselves were almost not degraded within 60 minutes by comparing lane 1 and 2, lane 12 and 13. Lane 3 showed the protein molecular weight distribution of protein plus inactivated pepsin. The protein bands at 120,72,50,20,15 kDa were visible as dominant bands but were markedly reduced at 0.5 min of digestion and barely visible through most time compared to line 11(1/5 loading amount). The protein bands had almost completely disappeared in lane 5 (2min), indicating that the tested protein (FC) could be digested by more than 80% within 2 min in the simulated gastric fluid digestion process in vitro. However, some of the protein bands between 25-40 kDa molecular weight coincided with some pepsin bands.

Although the high molecular weight protein (~120 kDa) in Figure 2. could not be shown in lane 11 (20% protein residue), the absence of bands for this protein on lanes 4-10 could also indicate that the high molecular weight protein (~120 kDa) can be digested by pepsin.

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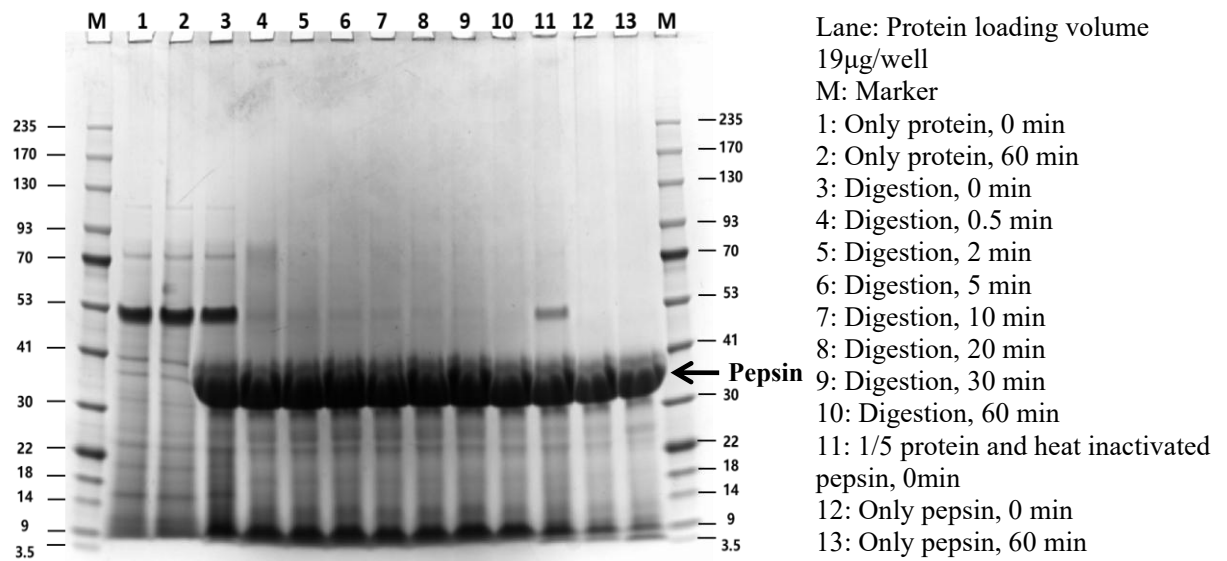


Figure 2. SDS-PAGE of test protein sample in E-SGF

## Part 2: The In Vitro Digestion of Protein Extracts (FC) using E-SIF

### 1. Reagents and Instruments

- (1) SIF (pH 7.5): 0.685g  $\text{KH}_2\text{PO}_4$  into 90mL distilled water, adjusted pH to 7.5 with approximately 2.5 mL of 2N NaOH solution, made the final volume to 100mL;
- (2) E-SIF (pH 7.5): 0.1g pancreatin (Maclin, cat#P916050) into 10mL SIF solution, preparing E-SIF (10 mg/ml), prepared as needed;
- (3) 5×SDS-PAGE Protein Loading Buffer (PLB): YEASEN, #20315ES;
- (4) Test protein (Fusarium compactum protein extract, 13.25mg/ml) and control protein (RUBISCO, 2mg/ml);
- (5) 15-well 4-20% High-Resolution Precast Gel (Bis-Tris): YEASEN, 36270ES10;
- (6) RealBand Pre-stained Protein Marker, 3 colors, Wide Range, 3.5~235 kDa for tris-mops-SDS running buffer, Sangon Biotech, cat#C610016;
- (7) Bio-Rad PowerPac Basic Power Pack;
- (8) ChampGel 5000 Gel Image Analyzer;
- (9) Gel Imaging Instrument: ChampGel 5000 Type Gel Imaging Instrument;
- (10) Gel Staining Solution: Dissolved 1.25g Coomassie Brilliant Blue R-250 in 225ml methanol, 50ml acetic acid and 225 water;
- (11) Destaining solution for Coomassie Brilliant Blue R-250: H<sub>2</sub>O, methanol, and acetic acid in a ratio of 65/25/10 (v/v/v);
- (12) HWS-26 Type Electric Heating Water Bath: Shanghai Yiheng Scientific Instrument Co., Ltd.;
- (13) SY-1230 Constant Temperature Water Bath: Suzhou Jiemei Electronics Co., Ltd.

### 2. Experimental Methods

#### 2.1 Preparation before experiment

- (1) Preheated two water baths to 37°C and 85°C;
- (2) Prepared SIF and E-SIF, incubate in advance in the water bath. Pancreatin (Maclin, cat#P916050) contains a variety of enzymes, including protease, amylase, lipase, etc.

#### 2.2 Reaction tubes

- (1) Labeled three 2ml centrifuge tubes as Pmix, Emix, Dmix;
- (2) Labeled thirteen 1.5mL tubes as P1/10, P0, P120, D0, D2, D5, D10, D20, D30, D60, D120, E0, E120, and added 10μl PLB to each tube.

#### 2.3 Protein Solution

- (1) P tube: Thawed the protein extract sample to room temperature, pipette an appropriate amount of solution and marked as P tube (13.25mg/ml);

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- (2) Pd tube: Pipetted a portion of the solution from the P tube, diluted the concentration to 1/10, and marked as Pd (10 $\mu$ L P + 90 $\mu$ L water);
- (3) RUBISCO solution: Prepared a 2mg/ml RUBISCO solution.

### 2.4 Preparation of digestion, protein, and enzyme controls.

- (1) Protein solution, pancreatin solution, and water were preheated separately for 5 minutes in 37°C before the start of each experiment;
- (2) P1/10: Added 38  $\mu$ L E-SIF, quickly heated to 85°C for 5 minutes, pipette 10  $\mu$ L of protein solution from the Pd tube, mixed well, heated at 85°C for 5 minutes, and stored on ice;
- (3) D0: Added 38  $\mu$ L E-SIF into two 1.5mL centrifuge tubes, quickly heated to 85°C for 10 minutes, pipetted 10  $\mu$ L of protein solution from the P tube or RUBISCO solution, mixed well, heated at 85°C for 5 minutes, and stored on ice;

- (4) Pmx (protein control without pancreatin): Pipetted 50 $\mu$ L from the P tube or RUBISCO solution, added 190 $\mu$ L SIF and mixed well;

P0: Pipetted 40 $\mu$ L from Pmx, mixed well, heated at 85°C for 5 minutes, and stored on ice;

P120: Heated the remaining Pmx at 37°C for 120 minutes, pipetted 40 $\mu$ L into P120, mixed well, heated at 85°C for 5 minutes, and stored on ice;

- (5) Emx (pancreatin control without protein): Pipetted 50 $\mu$ L of water, added 190 $\mu$ L E-SIF and mixed well;

E0: Pipetted 40 $\mu$ L from Emx, mixed well, heated at 85°C for 5 minutes, and stored on ice;

E120: Heated the remaining Emx at 37°C for 120 minutes, pipetted 40 $\mu$ L into E120, mixed well, heated at 85°C for 5 minutes, and stored on ice;

- (6) Dmx (digestion mixture): Pipetted 100 $\mu$ L from the P tube or RUBISCO solution, added 380 $\mu$ L E-SGF into two micro centrifuge tubes and mixed well, placed Dmx in 37°C water bath, pipetted 40 $\mu$ L of the digestion mixture at 2, 5, 10, 20, 30, 60, 120 minutes into D2-D120 tubes respectively, immediately heated each tube to 85°C for 5 minutes, and stored on ice.

### 2.5 Electrophoresis conditions

|   |    |     |    |    |    |     |     |     |     |      |       |    |      |    |
|---|----|-----|----|----|----|-----|-----|-----|-----|------|-------|----|------|----|
| 1 | 2  | 3   | 4  | 5  | 6  | 7   | 8   | 9   | 10  | 11   | 12    | 13 | 14   | 15 |
| M | P0 | P60 | D0 | D2 | D5 | D10 | D20 | D30 | D60 | D120 | P1/10 | E0 | E120 | M  |

(Sample loading volume: 10  $\mu$ L/well)

The digested samples were boiled for 5min and centrifuged at 4°C for 5min before adding into gel well;

Gel type: 4-20% High-Resolution Pre-cast Gel (Bis-Tris);

Marker (M): 3.5-235 kDa;

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Running Buffer: Tris-MOPS-SDS, 1 ×;

Electrophoresis conditions: 120V, about 60-80 minutes;

Fixing, staining, and destaining procedure: Staining with Coomassie Brilliant Blue R-250 staining solution for 10 min, decolorization overnight (at least 12 hours), then photographed using gel imaging system.

| Lane | Description                                   | Incubation time |     |
|------|-----------------------------------------------|-----------------|-----|
| M    | Marker (3.5-235 kDa)                          | NA              |     |
| 1    | protein control without pancreatin, (P0)      | 0               | min |
| 2    | protein control without pancreatin, (P120)    | 120             | min |
| 3    | protein in E-SIF, (D0)                        | 0               | min |
| 4    | protein in E-SIF, (D2)                        | 2               | min |
| 5    | protein in E-SIF, (D5)                        | 5               | min |
| 6    | protein in E-SIF, (D10)                       | 10              | min |
| 7    | protein in E-SIF, (D20)                       | 20              | min |
| 8    | protein in E-SIF, (D30)                       | 30              | min |
| 9    | protein in E-SIF, (D60)                       | 60              | min |
| 10   | protein in E-SIF, (D120)                      | 120             | min |
| 11   | 10% protein with quenched pancreatin, (P1/10) | 0               | min |
| 12   | Experimental control pancreatin, (E0)         | 0               | min |
| 13   | Experimental control pancreatin, (E120)       | 120             | min |
| M    | Marker (3.5-235 kDa)                          | NA              |     |

### 3. Results

3.1 The electrophoresis result of the simulated intestinal fluid digestion test sample was shown in Figure 3. The protein only without enzymes wells (lanes 1,2 of Figure 3) shows protein bands were similar with lanes 1,2 of Figure 2 in the SGF gel. The protein and enzyme themselves are almost not degraded within 120 minutes by comparing lanes 1 and 2, 11 and 12. Lane 3 showed the protein molecular weight distribution of all proteins and pancreatin. According to lane 4-10, the protein bands near 120, 72, 40, 35, 32, 20 and 15kDa (lanes 1,2 and 3 of Figure 3) had mostly disappeared at 2 min. Despite overlap between the ~53 kDa band and pancreatin control, >90% digestion at 2 min is evident from band intensity attenuation. These results indicate that pancreatin was able to fully digest the mixture of proteins from *Fusarium compactum*. Unfortunately, the protein bands near 25-30kDa overlapped with some bands of the pancreatin itself.

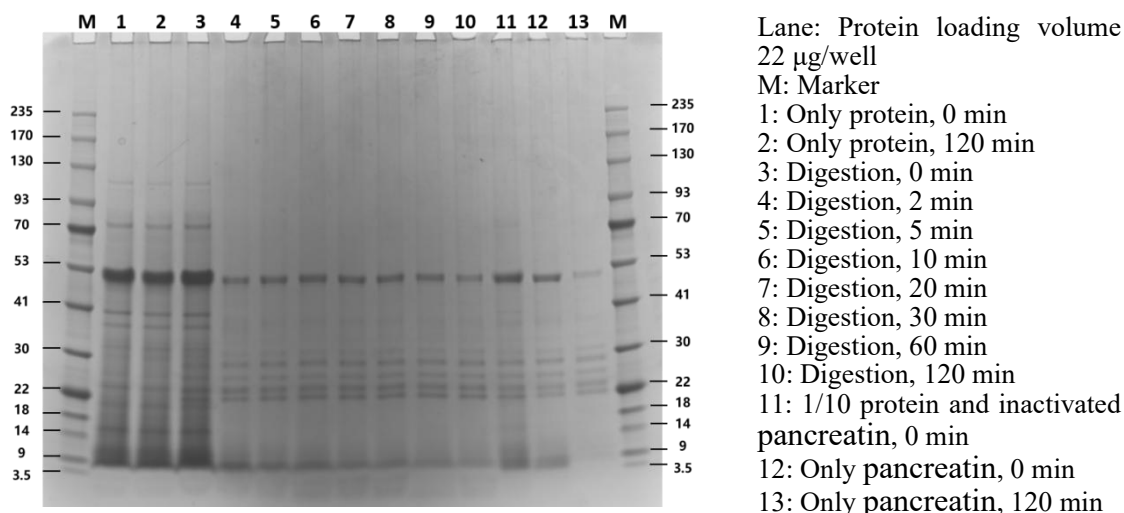


Figure 3. Test protein sample simulated intestinal fluid digestion SDS-PAGE

3.2 The electrophoresis result of the simulated intestinal fluid digestion of the RUBISCO sample was shown in Figure 4.

Figure 4 shows that the RUBISCO sample had two distinct bands near 50 and 12kDa. Although the 50kDa band overlapped with the pancreatin enzyme band, the 12kDa band has disappeared in lane 4 (2 minutes). It indicated that the activity of pancreatin met the requirements of digestion and RUBISCO was an unstable protein for intestinal simulated digest, used as a control protein which is easily digested in an SIF assay compared with the tested protein.

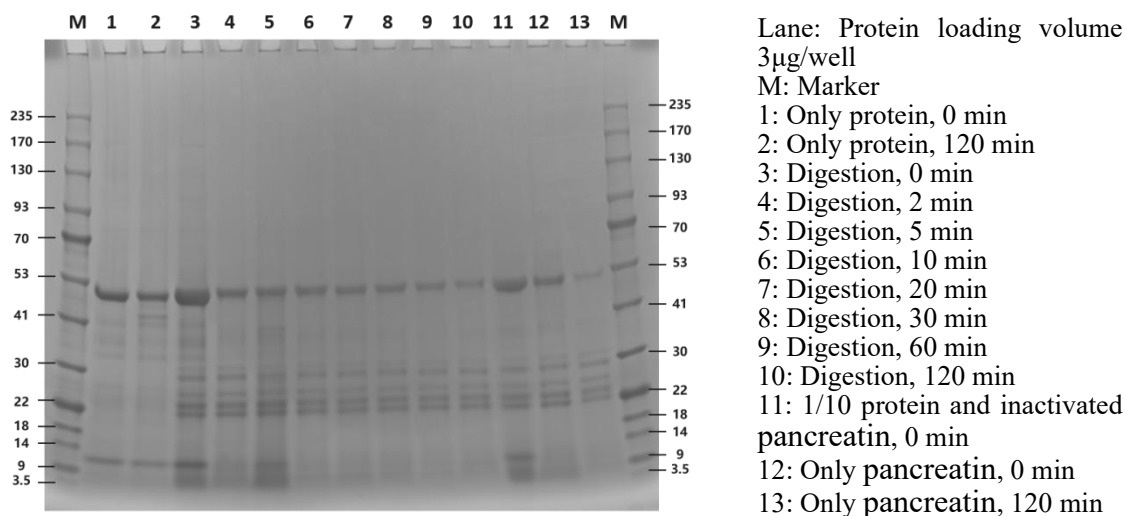


Figure 4. RUBISCO sample simulated intestinal fluid digestion SDS-PAGE

## Conclusion

The mycoprotein of *Fusarium compactum* produced by MoreMeat (Guangzhou) Biotech Co.,Ltd. via fermentation method could be rapidly digested by simulated gastric digestion, and the whole proteins do appear to be rapidly digested by simulated intestinal digestion. The results indicate that proteins from *Fusarium compactum* are relatively labile to enzymes from the human digestive tract. Therefore, it is less likely to cause allergic reactions in humans. Since some people may likely have allergies to the *Fusarium compactum* protein, digestion of the proteins is unlikely to alleviate all

risks.

## References

Astwood, J. D., Leach, J.N., and Fuchs, R. L. (1996). Stability of food allergens to digestion in vitro. *Nature Biotech.* 14: 1269-1273.

Bannon, G.A., Goodman, R.E., Leach, J.N., Rice, E., Fuchs, R.L., Astwood, J.D. (2002). Digestive stability in the context of assessing the potential allergenicity of food proteins. *Comments Toxicol.* 8: 271-285.

Codex Alimentarius (2003). Codex Alinorm 03/34: Joint FAO/WHO Food Standard Programme, Codex Alimentarius Commission, Twenty-Fifth Session, Rome, Italy 30 June-5 July, 2003.

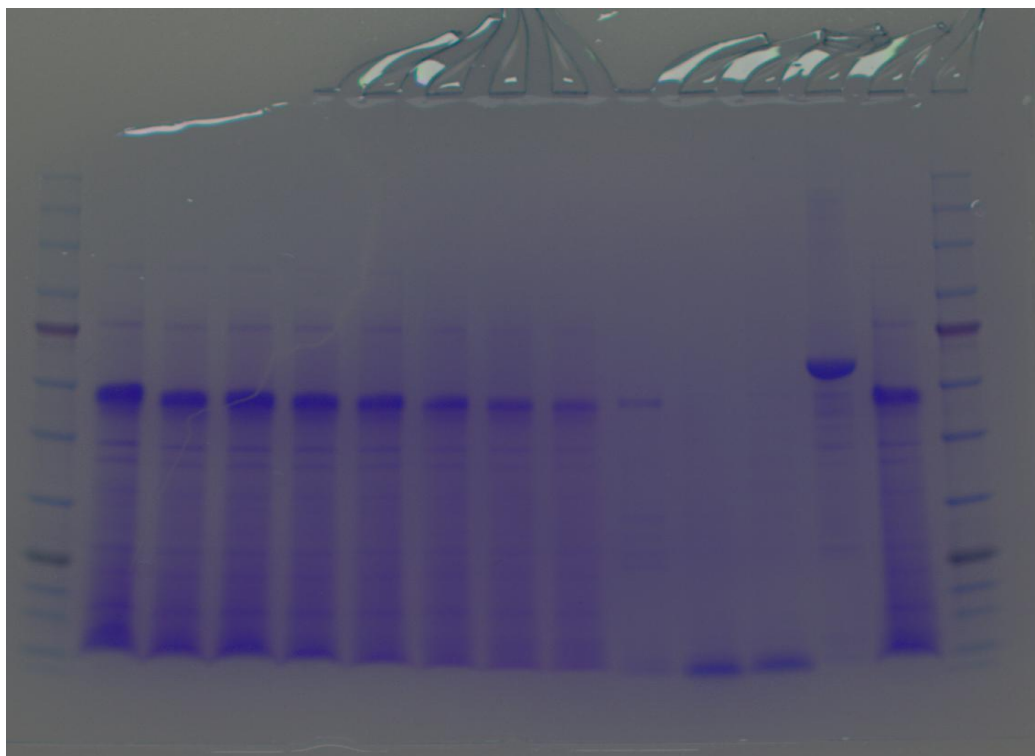
Thomas, K., Aalbers, M., Bannon, G.A., Bartels, M., Dearman, R.J., Esdaile, D.J., Fu, T.J., Glatt, C.M., Hadfield, N., Hatzos, C., Hefle, S.L., Heylings, J.R., Goodman, R.E., Henry, B., Herouet, C., Holsapple, M., Ladics, G., Landry, T.D., MacIntosh, S.C., Rice, E.A., Privalle, L.S., Steiner, H.Y., Teshima, R., van Ree, R., Woolhiser, M., Zwadny, J. (2004). A multi-laboratory evaluation of a common in vitro pepsin digestion assay protocol used in assessing the safety of novel proteins. *Regul. Toxicol. Pharmacol.* 39: 87-98.

Yu, H.-L., Ruan, W.-W., Cao, M.-J., Cai, Q.-F., Shen, H.-W., Liu, G.-M. (2013). Identification of physicochemical properties of *Scylla paramamosain* allergen, arginine kinase. *J. Sci. Food Agric.* 93: 245-253.

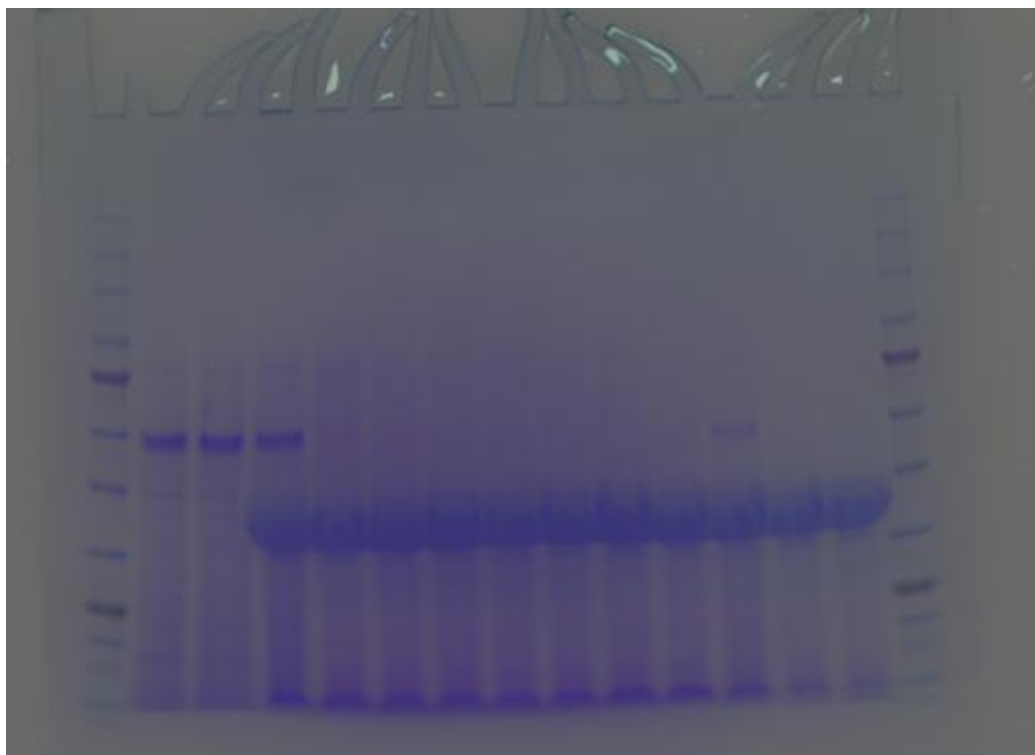
Agricultural and Rural Ministry Announcement No. 423-1-2021, Method for the Digestion Stability Test of Exogenous Proteins in Simulated Gastric and Intestinal Fluids for Food Safety Detection of Genetically Modified Organisms and Their Products [S].

Huang, Z.B., Tang, T., Gu, X., et al. (2023). Research progress on the assessment of food allergen allergenicity using an in vitro digestion model. *Food Research and Development*, 44(16): 213-219.

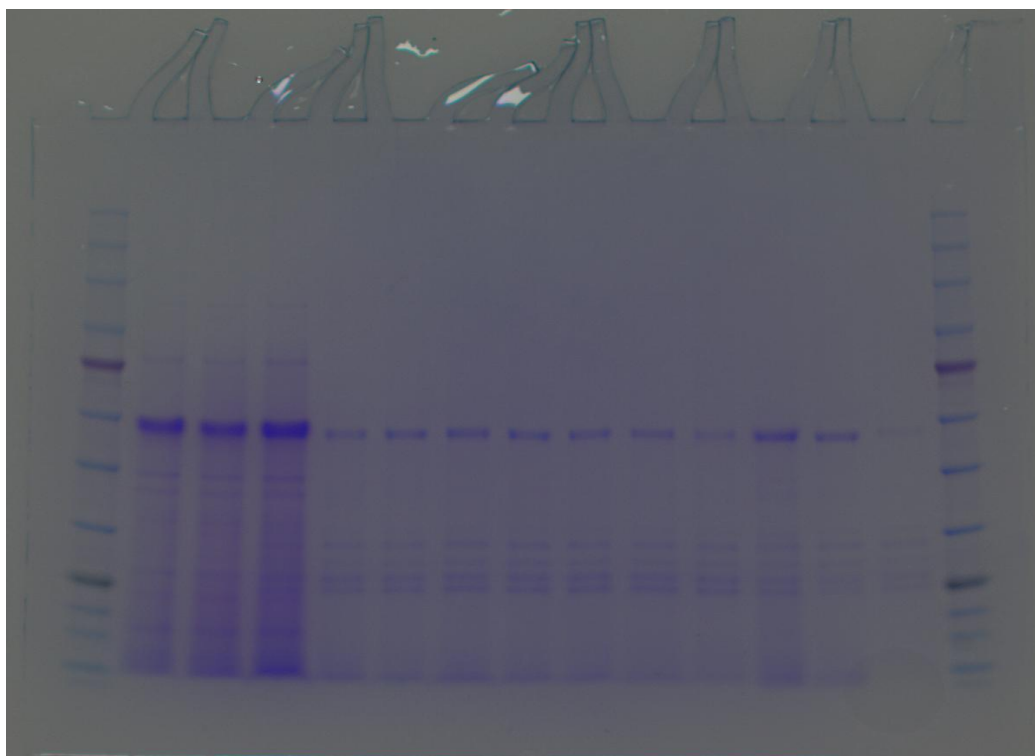
**Attachment**



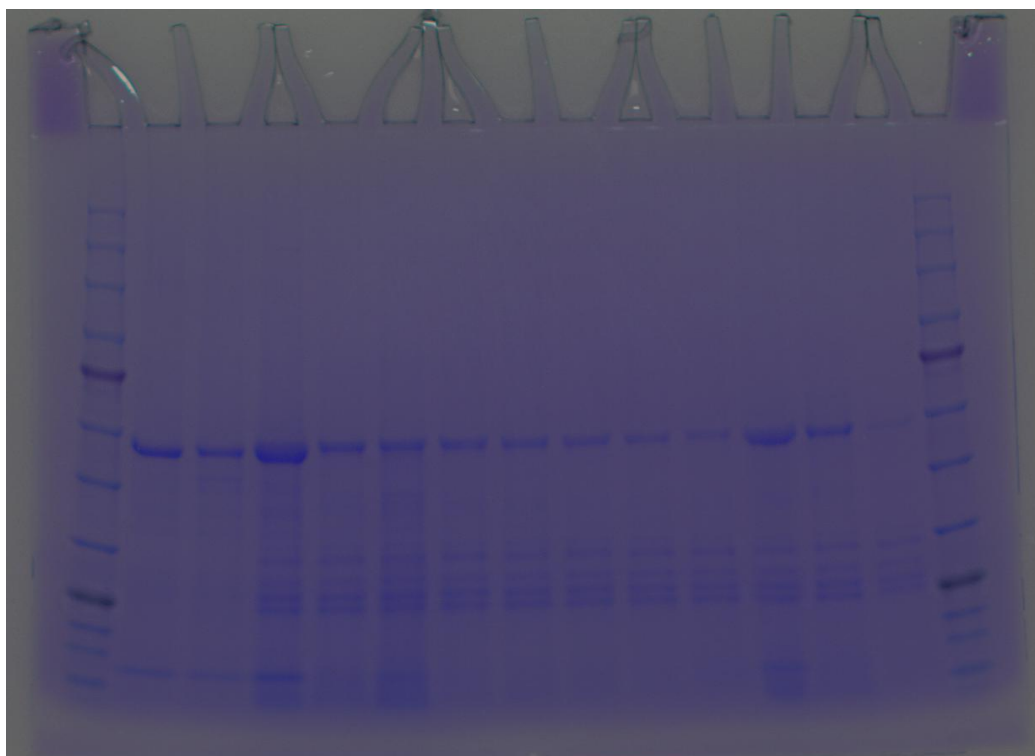
**Figure 5. Original picture of SDS-PAGE Gel showing the serial dilution of *Fusarium compactum* proteins**



**Figure 6. Original SDS-PAGE gel picture of test protein sample in E-SGF**



**Figure 7. Original SDS-PAGE gel picture of test protein sample in E-SIF**



**Figure 8. Original SDS-PAGE gel picture of RUBISCO in E-SIF**

Kaiping Deng, Ph.D.  
Division of Food Ingredients  
Office of Pre-Market Additive Safety  
Human Foods Program

Re: GRAS Notice No. GRN 001255

Dear Dr. Deng:

Per your request, Anchor Center for Certification (ACC), acting as the agent for MoreMeat (Guangzhou) Biotech Co., Ltd (MoreMeat), is providing a 2<sup>nd</sup> round of responses to FDA's questions as denoted in your email dated August 21, 2025, as follows:

- In the amendment dated July 25, 2025, the data point at 12 months in Table 2 is missing for lot MM134254009. Please provide the missing data point or a discussion of why this point was not included.

MoreMeat's response:

The stability study at long-term condition ( $-20^{\circ}\text{C}\pm 5^{\circ}\text{C}$ ) is still ongoing for lot MM13524009, therefore, the 12-month results are not presented in the table. With the completed stability results from lots MM13524002 and MM13524004, together with the trend data available from lot MM13524009, MoreMeat expects the 12-month results to remain within product specifications.

- In the amendment dated July 25, 2025, the notifier stated that mycelial biomass of *Fusarium compactum* "MM-135" (fungal protein) is intended to be used in only a subset of the general food categories provided in Table 13 as specified by the food codes provided in Appendix 2. We note that the food categories presented in Table 13 include uses for which food codes have not been provided in Appendix 2. We have modified Table 13 to include only those food uses in which there were food codes provided in Appendix 2. Please confirm that the food uses provided in the modified Table A (see below) include all the intended uses for fungal protein. If there are additional uses for fungal protein not shown in the table below, please provide a revised table of the intended uses and a revised dietary exposure estimate that includes these food uses.
  - If a revised dietary exposure estimate is completed and estimated dietary exposure is increased compared to the initial notice or amendment dated July 25, 2025, please provide updated calculations and related discussions regarding the safety of exposure to the ingredient, including protein, fiber, and mycotoxin levels.

MoreMeat's response:

MoreMeat accepts FDA's modification to Table 13, as shown in Table A, to include only those food uses in which there were food codes provided in Appendix 2.

- It is not clear if an analysis for aflatoxin B1 was conducted on fungal protein. Table 10 of the amendment dated July 25, 2025 reports the results as "Not Reported" while Table 6 of the GRAS notice reports "Not Detected". Please clarify this discrepancy.

MoreMeat's response:

MoreMeat confirms that testing for aflatoxin B1 was conducted on our mycoprotein product, and the results were 'not detected'. The results presented in Table 10 of the amendment dated July 25, 2025, were incorrect and should be corrected to 'not detected'.

- In the amendment dated July 25, 2025, the notifier's response to question 7 includes Table 4 which shows the percentage of genomic similarity between *F. compactum* "MM-135", *F. venenatum* A 3/5, and *F. flavolapis* MK7. However, the discussion related to this question does not discuss any differences in the proteome or metabolome between the strains. Similar product specifications, such as protein and fiber levels, or similar nutritional and amino acid profiles do not adequately justify safety of an ingredient. Please discuss any proteomic or metabolomic differences between the strain that is the subject of the notice and the strains used to support the safety narrative, and address why those differences do not pose a safety concern.

MoreMeat's response:

The proteomic differences between MoreMeat's mycoprotein product and the substances of GRN 000904 and GRN 000945 primarily focus on the potential allergenicity of the three fungal strains. As outlined in our GRAS dossier, analysis of MoreMeat's mycoprotein (derived from *F. compactum*) using the online database AllergenOnline ([www.allergenonline.org](http://www.allergenonline.org)) indicates that the product is not expected to pose a risk for food allergy. This conclusion is further supported by human consumption data, which did not report any adverse effects.

As for the metabolomic differences among the three strains, our assessment primarily focuses on mycotoxins, which are secondary metabolites produced by *Fusarium* species and may pose safety concerns for human consumption. As stated in our responses to Question 11 in the amendment dated July 25, 2025, we acknowledge that different mycotoxins may be present in the three *Fusarium* strains. Even though the metabolic pathways involved in mycotoxin biosynthesis are similar for *Fusarium* species, the levels of mycotoxins in mycoprotein derived from different strains can vary. Based on the literature (Nesic, 2014)<sup>1</sup>, both *F. compactum* and *F. venenatum* possess the genetic potential to encode trichothecenes; however, analyses of the respective mycoprotein products have consistently shown that trichothecenes are not detectable. We therefore conclude that any such compounds, if present, are below the limits of detection.

In addition, mycotoxins are not retained within fungal cells; once produced, they are secreted into the fermentation medium. Unlike measurements in contaminated

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<sup>1</sup> Nesic K, Ivanovic S, Nesic V. Fusarial toxins: secondary metabolites of *Fusarium* fungi. *Rev Environ Contam Toxicol*. 2014;228:101-20. doi: 10.1007/978-3-319-01619-1\_5. PMID: 24162094.

grains reported in the literature (Nesic, 2014), in MoreMeat's mycoprotein production process, the mycotoxins are released into the fermentation broth and largely removed in the spent media. Consequently, only minimal residual levels are present in the final mycoprotein product, providing an inherent safety assurance. These contribute to the observed differences in mycotoxin levels between the strain that is the subject of this notice and the strains used to support the safety narrative.

However, despite the metabolomic differences among the three strains, these differences are not expected to pose any safety concern. Any mycotoxins potentially present in MoreMeat's mycoprotein derived from *F. compactum* are controlled through established limits in the product specifications to ensure safety.

- Please provide data and information on human consumption and/or tolerability of fungal protein and discuss if its consumption resulted in any adverse gastrointestinal effects or similar tolerability concerns.

MoreMeat's response:

MoreMeat's mycoprotein product has been consumed by thousands of consumers since 2022 without any reports of tolerability issues or adverse effects. In addition, MoreMeat has participated in various exhibition tasting events and conducted hundreds of internal tastings, with no reported concerns regarding tolerability.

MoreMeat conducted a tasting survey with 220 consumers in China. The survey participants were 20–60 years old, comprising 40% females and 60% males. The mycoprotein product was served in the form of meatballs (containing 45%-55% of MoreMeat's mycoprotein) or non-dairy milk (containing 10-15% of MoreMeat's mycoprotein). All participants completed an online questionnaire. MoreMeat's mycoprotein product was well accepted as a meat substitute, with no adverse effects reported.

Furthermore, MoreMeat has exhibited at several conferences, where thousands of consumers were served dumplings, pizza, meatballs, and other meat substitutes prepared with MoreMeat's mycoprotein. No tolerability concerns were observed.

If additional information or clarification is needed as you and your colleagues proceed with the review, please feel free to contact me via email.

Sincerely,  
Stella Si  
Anchor Center for Certification (ACC)  
No. 1295 Chuan Qiao Road, Building 2, Suite 302  
Shanghai, 201206  
CHINA

Table A. Revisions to the intended uses based on the food codes provided in Appendix 2 of the submitted GRAS Notice.

| <b>Food Category</b>                         | <b>Food Uses</b>                                                                                                                       | <b>Maximum use level (%)</b> |
|----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| <b>Baked Goods</b>                           | Breads, rolls, bagels, muffins, biscuits, cornbread, tortillas, cakes, cookies, pies, pastries, doughnuts, crackers, pancakes, waffles | 50%                          |
| <b>Dairy Analogs</b>                         | Non-dairy milk (soy, almond, rice, oat, coconut), coffee creamers, imitation cheese, nutritional drinks and shakes                     | 50%                          |
| <b>Non-dairy Yogurt</b>                      | Coconut milk yogurt, almond milk yogurt                                                                                                | 70%                          |
| <b>Fats and Oils, Condiments and Sauces</b>  | Cheese sauce, fish sauce, gravies, soy-based sauces, condiments (ketchup, mustard, hot sauce), salad dressings, mayonnaise             | 30%                          |
| <b>Grain Products</b>                        | Pasta (regular, vegetable, whole grain, gluten-free), noodles (Chow Mein, rice noodles)                                                | 50%                          |
| <b>Meat and egg substitutes</b>              | Imitation crab, egg substitutes, soybean curd, meat substitutes (bacon, chicken, hot dogs, burgers), protein powder mixes              | 90%                          |
| <b>Processed Fruit and Fruit Juices</b>      | Fruit smoothies (with/without dairy), high protein juices                                                                              | 60%                          |
| <b>Processed Vegetable Juices</b>            | Tomato juice cocktail, mixed vegetable juice, vegetable smoothies                                                                      | 60%                          |
| <b>Soups and Soup Mixes</b>                  | Canned soups* (beef, chicken, bean, lentil, tomato, vegetable)                                                                         | 60%                          |
| <b>Frozen or Shelf Stable Prepared Meals</b> | Frozen dinners, pizza, pasta dishes, turnovers, shelf stable prepared meals                                                            | 60%                          |

\* Excluding use in soups under the jurisdiction of the United States Department of Agriculture

Kaiping Deng, Ph.D.  
Division of Food Ingredients  
Office of Pre-Market Additive Safety  
Human Foods Program

Re: GRAS Notice No. GRN 001255

Dear Dr. Deng:

Per your request, Anchor Center for Certification (ACC), acting as the agent for MoreMeat (Guangzhou) Biotech Co., Ltd (MoreMeat), is providing a 2<sup>nd</sup> round of responses to FDA's questions as denoted in your email dated August 21, 2025, as follows:

- In the amendment dated July 25, 2025, the data point at 12 months in Table 2 is missing for lot MM134254009. Please provide the missing data point or a discussion of why this point was not included.

MoreMeat's response:

The stability study at long-term condition ( $-20^{\circ}\text{C}\pm 5^{\circ}\text{C}$ ) is still ongoing for lot MM13524009, therefore, the 12-month results are not presented in the table. With the completed stability results from lots MM13524002 and MM13524004, together with the trend data available from lot MM13524009, MoreMeat expects the 12-month results to remain within product specifications.

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  - If a revised dietary exposure estimate is completed and estimated dietary exposure is increased compared to the initial notice or amendment dated July 25, 2025, please provide updated calculations and related discussions regarding the safety of exposure to the ingredient, including protein, fiber, and mycotoxin levels.

MoreMeat's response:

MoreMeat accepts FDA's modification to Table 13, as shown in Table A, to include only those food uses in which there were food codes provided in Appendix 2.

- It is not clear if an analysis for aflatoxin B1 was conducted on fungal protein. Table 10 of the amendment dated July 25, 2025 reports the results as "Not Reported" while Table 6 of the GRAS notice reports "Not Detected". Please clarify this discrepancy.

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Furthermore, MoreMeat has exhibited at several conferences, where thousands of consumers were served dumplings, pizza, meatballs, and other meat substitutes prepared with MoreMeat's mycoprotein. No tolerability concerns were observed.

If additional information or clarification is needed as you and your colleagues proceed with the review, please feel free to contact me via email.

Sincerely,  
Stella Si  
Anchor Center for Certification (ACC)  
No. 1295 Chuan Qiao Road, Building 2, Suite 302  
Shanghai, 201206  
CHINA

Table A. Revisions to the intended uses based on the food codes provided in Appendix 2 of the submitted GRAS Notice.

| <b>Food Category</b>                         | <b>Food Uses</b>                                                                                                                       | <b>Maximum use level (%)</b> |
|----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| <b>Baked Goods</b>                           | Breads, rolls, bagels, muffins, biscuits, cornbread, tortillas, cakes, cookies, pies, pastries, doughnuts, crackers, pancakes, waffles | 50%                          |
| <b>Dairy Analogs</b>                         | Non-dairy milk (soy, almond, rice, oat, coconut), coffee creamers, imitation cheese, nutritional drinks and shakes                     | 50%                          |
| <b>Non-dairy Yogurt</b>                      | Coconut milk yogurt, almond milk yogurt                                                                                                | 70%                          |
| <b>Fats and Oils, Condiments and Sauces</b>  | Cheese sauce, fish sauce, gravies, soy-based sauces, condiments (ketchup, mustard, hot sauce), salad dressings, mayonnaise             | 30%                          |
| <b>Grain Products</b>                        | Pasta (regular, vegetable, whole grain, gluten-free), noodles (Chow Mein, rice noodles)                                                | 50%                          |
| <b>Meat and egg substitutes</b>              | Imitation crab, egg substitutes, soybean curd, meat substitutes (bacon, chicken, hot dogs, burgers), protein powder mixes              | 90%                          |
| <b>Processed Fruit and Fruit Juices</b>      | Fruit smoothies (with/without dairy), high protein juices                                                                              | 60%                          |
| <b>Processed Vegetable Juices</b>            | Tomato juice cocktail, mixed vegetable juice, vegetable smoothies                                                                      | 60%                          |
| <b>Soups and Soup Mixes</b>                  | Canned soups* (beef, chicken, bean, lentil, tomato, vegetable)                                                                         | 60%                          |
| <b>Frozen or Shelf Stable Prepared Meals</b> | Frozen dinners, pizza, pasta dishes, turnovers, shelf stable prepared meals                                                            | 60%                          |

\* Excluding use in soups under the jurisdiction of the United States Department of Agriculture